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Merchant terminal for receiving payment from a vehicle

Abstract

Methods, systems, apparatus, and non-transitory computer readable media are described for using a vehicle as a payment device. Various aspects may include receiving a selection of a stored financial card or financial account at a vehicle head unit. The selected financial card or financial account may be transmitted to a point-of-sale (POS) terminal for making a payment by transmitting a tokenized card number to the POS terminal. The tokenized card number may be transmitted over a very short-range communication link to ensure that the transmission is secure. For example, electronic circuitry may be attached to the exterior of the vehicle, where the electronic circuitry may be within a threshold distance (e.g. one inch, three inches, six inches, one foot, three feet, etc.) of the POS terminal. The tokenized card number may be transmitted from the vehicle head unit to the electronic circuitry and then to the POS terminal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/168,001 entitled “Merchant Terminal for Receiving Payment from a Vehicle” filed on Feb. 4, 2021, which is a continuation of U.S. patent application Ser. No. 15/428,656 entitled “Merchant Terminal for Receiving Payment from a Vehicle,” filed on Feb. 9, 2017, which claims priority to and the benefit of the filing date of (1) provisional U.S. Patent Application No. 62/295,596 entitled “Connected Car as a Payment Device,” filed on Feb. 16, 2016, (2) provisional U.S. Patent Application No. 62/360,015 entitled “Merchant Terminal for Receiving Payment from a Vehicle,” filed on Jul. 8, 2016, and (3) provisional U.S. Patent Application No. 62/362,152 entitled “Merchant Terminal for Receiving Payment from a Vehicle,” filed on Jul. 14, 2016, the entire contents of each of which is hereby expressly incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure generally relates to systems, methods, apparatus, and non-transitory computer readable media for using a vehicle as a payment device and, more particularly to completing a financial transaction by communicating from a vehicle to a point-of-sale (POS) terminal.

BACKGROUND

(2) Today, vehicle occupants pay for several goods and services while the vehicle occupants are inside or near their vehicles, such as food, fuel, a car wash, etc. Typically, the transaction takes place by swiping a financial card such as a credit or debit card at a POS terminal or by providing cash or the financial card to an employee of the establishment which provides the goods or services. However, this requires a vehicle occupant to constantly carry cash and/or financial cards when making purchases, get out of her vehicle to make payments, and/or exchange cash or a financial card with an employee, which may be a time-consuming process.

BRIEF SUMMARY

(3) The present embodiments may relate to using a vehicle as a payment device, so that users may remain

inside their vehicles when making payments for goods and services, such as food, fuel, a car wash, etc. An application may be stored within an infotainment system (also referred to herein as a “vehicle head unit”) of a vehicle which stores data representing financial cards for making payments. To retrieve one of the stored financial cards for making a payment at a drive-thru, fuel station, etc., the user may select an indication of a financial card at the infotainment system.

(4) In response to the user's selection, the infotainment system may transmit the selected data over a short-range communication link to a POS terminal at the drive-thru, fuel station, etc. The POS terminal may then process the payment using the transmitted data and/or transmit an electronic receipt to be displayed on the infotainment system. In some embodiments, for additional security and/or to establish a communication link between the infotainment system and the POS terminal, electronic circuitry may be proximately attached to the exterior of the vehicle, such as at or near the fuel tank cap (also referred to herein as a “fuel cap”). The electronic circuitry may receive the selected data from the infotainment system over a first short-range communication link, and/or transmit the received data to the POS terminal over a second very short-range communication link.

(5) In one aspect, a computer-implemented method for using a point-of-sale terminal to accept vehicle payments may be provided. The method may include: (1) receiving (via one or more processors and/or associated transceivers in a POS terminal) an indication that a vehicle for providing electronic payments is within a predetermined range of the POS terminal; (2) transmitting (via the one or more processors and/or associated transceivers, and/or a wired or wireless communication and/or data transmission over a radio link or wireless communication channel) an electronic message including an indication of goods or services that can be purchased at the POS terminal and a user control for transmitting a request to order the goods or services to the POS terminal, wherein the electronic message is displayed on a head unit of the vehicle; in response to receiving an indication from the vehicle that the vehicle or a user within the vehicle wants to initiate a payment process; and/or (3) receiving (via the one or more processors and/or associated transceivers, and/or a wired or wireless communication and/or data transmission from an electronic device proximately attached to an exterior of the vehicle over a radio link or wireless communication channel) information indicative of a financial account for making payments. The method may further include: (4) providing (via the one or more processors) the goods or services to the vehicle; (5) transmitting (via the one or more processors and/or associated transceivers, and/or a wired or wireless communication and/or data transmission over a radio link or wireless communication channel) the information indicative of the financial account to an issuing bank server for processing payment for the goods or services; and/or when the payment is accepted as indicated by the issuing bank server, (6) transmitting (via the one or more processors and/or associated transceivers, and/or a wired or wireless communication and/or data transmission to the vehicle over a radio link or wireless communication channel) an electronic receipt indicative of the payment to facilitate using the vehicle as a payment device. The method may include additional, fewer, or alternative actions, including those discussed elsewhere herein.

(6) In another aspect, a merchant communication terminal for facilitating using a vehicle as a payment device may be provided. The merchant communication terminal may include one or more processors (and/or associated transceivers) in a merchant communication terminal associated with a merchant, and/or a non-transitory computer-readable memory coupled to the one or more processors and storing machine readable instructions, that when executed by the one or more processors, may cause the merchant communication terminal to perform various tasks. For example, the instructions may cause the system to: (1) detect a triggering event indicating that a vehicle is within a predetermined range for communicating with the merchant communication terminal; (2) establish a short-range communication link between the merchant communication terminal and the vehicle; (3) electronically verify identification information indicative of the vehicle or a user within the vehicle; (4) electronically verify a virtual account associated with the vehicle or the user within the vehicle; (5) accept a virtual order for goods or services provided by the merchant; and/or (6) automatically provide the goods or services to the vehicle to facilitate using the vehicle as a payment device. The system may include additional, fewer, or alternate components and/or functionality, including that discussed elsewhere herein.

(7) In yet another aspect, a merchant communication terminal for using an autonomous vehicle as an electronic payment device may be provided. The merchant communication terminal may include one or more processors (and/or associated transceivers) in a merchant communication terminal associated with a merchant, and/or a non-transitory computer-readable memory coupled to the one or more processors and storing machine readable instructions, that when executed by the one or more processors, may cause the

merchant communication terminal to perform various tasks. For example, the instructions may cause the system to: (1) obtain identification information for an autonomous vehicle; (2) verify that the autonomous vehicle has not been reported stolen by communicating the identification information to a server for retrieving stolen vehicle records; (3) detect a triggering event indicating that the autonomous vehicle is within a predetermined range for communicating with the merchant communication terminal; and/or (4) receive, via a short-range communication link, virtual account information for a virtual account associated with the vehicle. The instructions may further cause the system to: (5) receive and accept, via the short-range communication link, an input from the vehicle authorizing payment for goods and services provided by the merchant; (6) securely receive, via the short-range communication link, funds from the virtual account at a virtual account for the merchant; and/or (7) transmit, via the short-range communication link, a notification that electronic payment for the goods or services has been accepted to facilitate the autonomous vehicle acting as a payment device.

(8) Advantages will become more apparent to those skilled in the art from the following description of the preferred embodiments which have been shown and described by way of illustration. As will be realized, the present embodiments may be capable of other and different embodiments, and their details are capable of modification in various respects. Accordingly, the drawings and description are to be regarded as illustrative in nature and not as restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The figures described below depict various aspects of the system and methods disclosed therein. It should be understood that each figure depicts an embodiment of a particular aspect of the disclosed system and methods, and that each of the figures is intended to accord with a possible embodiment thereof. Further, wherever possible, the following description refers to the reference numerals included in the following figures, in which features depicted in multiple figures are designated with consistent reference numerals.

(2) There are shown in the drawings arrangements which are presently discussed, it being understood, however, that the present embodiments are not limited to the precise arrangements and instrumentalities shown, wherein.

(3) FIG. 1 illustrates a block diagram of an exemplary environment on which an exemplary vehicle payment system may operate in accordance with an exemplary aspect of the present disclosure;

(4) FIG. 2A illustrates a block diagram of an exemplary vehicle head unit in accordance with an exemplary aspect of the present disclosure;

(5) FIG. 2B illustrates a block diagram of exemplary electronic circuitry attached to the exterior of the vehicle in accordance with an exemplary aspect of the present disclosure;

(6) FIG. 3A depicts an exemplary financial card entry screen of a vehicle payment application in accordance with an exemplary aspect of the present disclosure;

(7) FIG. 3B depicts an exemplary activation screen of a vehicle payment application in accordance with an exemplary aspect of the present disclosure;

(8) FIG. 3C depicts an exemplary financial card selection screen of a vehicle payment application in accordance with an exemplary aspect of the present disclosure;

(9) FIG. 3D depicts an exemplary payment authorization screen of a vehicle payment application in accordance with an exemplary aspect of the present disclosure;

(10) FIG. 3E depicts an exemplary electronic receipt screen of a vehicle payment application in accordance with an exemplary aspect of the present disclosure;

(11) FIG. 4 depicts a flow diagram representing an exemplary method for implementing the vehicle payment system in accordance with an exemplary aspect of the present disclosure;

(12) FIG. 5 depicts an exemplary connected vehicle configured for use as a payment device in accordance with an exemplary aspect of the present disclosure;

(13) FIG. 6 depicts an exemplary system that is configured to facilitate using a connected vehicle as a payment device in accordance with an exemplary aspect of the present disclosure;

(14) FIG. 7 depicts an exemplary system that is configured to facilitate using a connected vehicle as a payment device including a robotic arm in an extended position in accordance with an exemplary aspect of the present disclosure; and

(15) FIG. 8 depicts a flow diagram representing an exemplary computer-implemented method of conducting commercial transactions with an autonomous vehicle configured for vehicle pay via merchant terminals in accordance with an exemplary aspect of the present disclosure.

(16) The figures depict preferred embodiments for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the systems and methods illustrated herein may be employed without departing from the principles of the invention described herein.

DETAILED DESCRIPTION

(17) To utilize a vehicle as a payment device, a vehicle payment application may be included in a vehicle head unit. The vehicle payment application may store one or more financial cards or any other financial accounts (also referred to herein as a “virtual account”) of a user, which may be retrieved for making payments from the vehicle. The financial cards or financial accounts may be received at the vehicle payment application by, for example, scanning an image of the financial card at the vehicle head unit, entering data for the financial card or financial account (also referred to herein as “virtual account information”) such as a cardholder name, card issuer, card number, card expiration date, etc., and/or transferring data for the financial card or financial account from a mobile device such as a smart-phone, wearable device, laptop computer, tablet, etc.

(18) In some embodiments, the vehicle payment application may generate a token representing the financial data for the financial card (also referred to herein as a “tokenized card number”) and/or may store the token rather than the financial data to reduce the risk of a security breach. The financial data may include a cardholder name, a card number, a card expiration date, a card security code (CSC), and/or a card type (e.g., VISA®, MasterCard®, American Express®, Discover®, etc.). In some embodiments, the financial data may also include additional information associated with the financial card such as a billing address.

(19) Also in some embodiments, the financial data may be transmitted to a third-party token server which may generate and/or transmit the token to the vehicle head unit. The third-party token server may store the token with the corresponding financial data for the financial card. When the payment is processed, the issuing bank may communicate with the third-party token server to retrieve the financial data corresponding to the token, for example. In yet other embodiments, the third-party token server may be the issuing bank.

(20) In any event, when a user wants to make a purchase from the vehicle, the user may select a user control requesting to make a payment from the vehicle. The vehicle payment application may then authenticate the user to ensure that the user is authorized to make payments with the stored financial cards or financial accounts. For example, the user may be authenticated by receiving biometric information from the user such as a fingerprint, an image of the user's face or eyes, etc., and comparing the biometric information to stored biometric information for authorized users. In other embodiments, the user may be authenticated by entering a username and/or password. In any event, when the user is authenticated, the user may select an indication of one of the stored financial cards at the vehicle head unit via the vehicle payment application. An indication of a financial card may be an image representing the financial card including the shape of the financial card, the background color/image of the financial card, the name of the issuer of the financial card (also referred to herein as an “issuing bank”), the card type, and/or masked financial data.

(21) A token representing financial data for the selected financial card may then be transmitted to a POS terminal at the establishment where the user is making the purchase. For example, the token may be transmitted to the POS terminal via a short-range communication link, such as Bluetooth, Universal Serial Bus (USB), Dedicated Short-Range Communications (DSRC), Radio-Frequency Identification (RFID), etc. In some embodiments, the vehicle payment system may include electronic circuitry (also referred to herein as an “electronic device”) proximately attached to the exterior of the vehicle, such as a door panel, fuel cap, etc. In such embodiments, the vehicle head unit may communicate with the electronic circuitry via a first short-range communication link, such as Bluetooth, for example. The electronic circuitry may in turn communicate with the POS terminal, via a second very short-range communication link such as Near Field Communication (NFC), high frequency (HF) RFID, etc.

(22) A short-range communication link may be a communication link between devices where data is transmitted via a wired and/or wireless connection within a first threshold distance (e.g. 30 feet, 50 feet, 100 feet, 200 feet, etc.). For example, a short-range communication link may include Bluetooth, DSRC, RFID, Wi-Fi, USB, etc. A very short-range communication link may be a communication link between devices where data is transmitted via a wired and/or wireless connection within a second threshold distance which is less than the first threshold distance (e.g., one inch, three inches, six inches, a foot, three feet, etc.). For example, a very short-range communication link may include NFC, HF RFID, etc. In some scenarios, very

short-range communication links may also be short-range communication links.

(23) For example, the vehicle head unit and the electronic circuitry may be paired together via a Bluetooth pairing. When the user requests to make a payment from the vehicle and selects a stored financial card or financial account, the vehicle head unit may transmit an indication to the electronic circuitry to transmit the token representing the financial data for the selected financial card. In turn, the electronic circuitry may transmit the token over the second very short-range communication link to the POS terminal.

(24) The electronic circuitry proximately attached to the exterior of the vehicle may be in very close proximity to the POS terminal (e.g. within one inch, three inches, six inches, a foot, three feet, etc.). By utilizing the electronic circuitry as an intermediary between the vehicle head unit and the POS terminal, the present embodiments advantageously allow for secure transmission of the token representing financial data. For example, by transmitting the token between devices which are within a few inches or feet of each other via a very short-range communication link, it becomes very difficult for eavesdroppers to intercept the signal. In this manner, the transmission may be secure while allowing the user to add and/or select financial cards, authorize payments, and/or view electronic receipts on the display of the vehicle head unit.

(25) Additionally, in some embodiments, the token may be transmitted from the vehicle head unit to the electronic circuitry once, when a new financial card or financial account is added. Subsequently, when the user authorizes payment for a selected financial card or financial account, an indication of the financial card or financial account may be transmitted to the electronic circuitry, which may in turn retrieve and/or transmit the token to the POS terminal. Accordingly, the present embodiments create additional security for the transmission of a token representing financial data, because the token is only transmitted from the vehicle head unit to the electronic circuitry once. After the initial transmission, all subsequent transmissions may be over very short distances, as mentioned above, making it difficult for eavesdroppers to intercept the signal.

(26) In addition to transmitting payments from a vehicle in a secure manner, the present embodiments advantageously allow for further automation of the retail process. For example, as described in more detail below, a vehicle at a fuel station may automatically transmit payment, open a fuel cap for receiving the fuel and interact with a smart fuel pump having a robotic arm to maneuver the nozzle and fill the vehicle with fuel. In this manner, autonomous vehicles may travel to the fuel station, receive fuel and make a payment without any human intervention.

(27) Generally speaking, the techniques for utilizing a vehicle as a payment device may be implemented in one or more network servers, in one or more client devices, in a vehicle head unit, in electronic circuitry attached to the exterior of the vehicle, or a system that includes several of these devices. However, for clarity, the examples below focus primarily on an embodiment in which financial data for a financial card is received at a vehicle head unit. The vehicle head unit may transmit the financial data to a third-party token server which may generate and/or transmit a token representing the financial data to the vehicle head unit. When a user selects the financial card for making a payment, the vehicle head unit may transmit the token representing financial data for the selected financial card to electronic circuitry attached to the exterior of the vehicle. The electronic circuitry may then transmit the token to the POS terminal for processing the payment. In other embodiments, the vehicle head unit may communicate directly with the POS terminal.

I. Exemplary Environment for Vehicle Payment Device

(28) FIG. 1 illustrates various aspects of an exemplary environment implementing a vehicle payment system **100**. The environment **100** may include a vehicle head unit **14** within a vehicle **12**, a portable device **10** within the vehicle **12**, electronic circuitry **50** proximately attached to the exterior of the vehicle, a third-party token server **104**, and/or a POS terminal **20** which may be communicatively connected through one or more short-range and/or very short-range communication links **120**, **122** and/or a network **130**, as described below.

(29) The vehicle **12** may be operated by a human, such as a user or may be an autonomous and/or semi-autonomous vehicle. An autonomous vehicle may be a vehicle including one or more automated operation features capable of operating the vehicle in the absence of or without operating input from a vehicle operator. A semi-autonomous vehicle may be a vehicle with one or more automated operation features capable of operating the vehicle without operating input from a vehicle operator and one or more manual operation features that require operating input from a vehicle operator.

(30) According to some embodiments, the vehicle head unit **14** may be a combination of hardware and software components, also as described in more detail below. The vehicle head unit **14** may include a display **18** for presenting application data. The display **18** in some implementations may be a touchscreen and may include a software keyboard for entering text input, such as financial data for a financial card, a financial card selection, etc. Hardware input controls **22** on the vehicle head unit **14** may be used for entering

alphanumeric characters or to perform other functions. The vehicle head unit **14** also may include audio input and output components such as a microphone and speakers, for example.

(31) Furthermore, the vehicle head unit **14** may communicate with a portable device **10** within the vehicle via a communication link, which may be wired (e.g., wired Universal Serial Bus (USB)) or wireless (e.g., Bluetooth, Wi-Fi Direct, wireless USB). For example, the portable device **10** may store indications of financial cards, tokens, and/or financial data which may be transmitted to the vehicle head unit **14** when a user adds a new financial card to the vehicle payment application. The portable device **10** may be a tablet computer, a cell phone, a personal digital assistant (PDA), a smart-phone, a laptop computer, a portable media player (not shown), a pager, a wearable computing device, smart glasses, smart watches or bracelets, phablets, other smart devices, devices configured for wired or wireless RF (Radio Frequency) communication, etc.

(32) Moreover, the vehicle head unit **14** may include one or more processor(s) such as a microprocessor coupled to a memory. The memory may be tangible, non-transitory memory and may include any types of suitable memory modules, including random access memory (RAM), read-only memory (ROM), flash memory, other types of persistent memory, etc. The memory may store, for example instructions executable on the processors for a vehicle payment application. The vehicle head unit **14** is described in more detail below with reference to FIG. 2A.

A. Exemplary Vehicle Payment Application

(33) To utilize a vehicle as a payment device, the vehicle payment application may receive and/or store indications of one or more financial cards or financial accounts, a token representing financial data for each financial card, and/or biometric or login information for authorized users to verify that the user is authorized to make payments with the stored financial cards. The vehicle payment application may present indications of each of the stored financial cards on the display **18** of the vehicle head unit **14** and one or more user controls for allowing the user to select one of the financial cards. Upon receiving a selection of one of the financial cards and a request to make payment, the vehicle payment application may transmit the token representing financial data for the selected financial data via a short-range communication link **120** to the POS terminal **20**. In some embodiments, the vehicle payment application may transmit the token via a first short-range communication link **120** to the electronic circuitry **50** proximately attached to the exterior of the vehicle. The electronic circuitry **50** may then transmit the token via a second very short-range communication link **122** to the POS terminal **20**. In any event, the POS terminal **20** may transmit an electronic receipt to the vehicle head unit **14**, and the vehicle payment application may present the electronic receipt on the display **18**.

B. Exemplary Electronic Circuitry

(34) The electronic circuitry **50** proximately attached to the exterior of the vehicle may be a combination of hardware and software components, also as described in more detail below. The electronic circuitry **50** may include a wireless transceiver for transmitting/receiving radio signals using NFC, Bluetooth, USB, DSRC, RFID, Wi-Fi, etc. The electronic circuitry **50** may communicate with the vehicle head unit **14** via a first short-range communication link **120**. The electronic circuitry **50** may also communicate with the POS terminal **20** via a second very short-range communication link **122**. In some embodiments, the electronic circuitry **50** may include one or more processor(s) such as a microprocessor coupled to a memory.

(35) Moreover, the electronic circuitry **50** may store tokens representing financial data for each financial card. For example, the electronic circuitry **50** may be paired with the vehicle head unit **14** via a Bluetooth connection. When the user enters a new financial card on the vehicle head unit, the generated token representing financial data for the financial card may be transmitted and/or stored in the electronic circuitry **50**. When the user later selects the financial card for making a payment, the vehicle head unit **14** may transmit an indication of the financial card to the electronic circuitry **50** and the electric circuitry **50** may retrieve the corresponding token and/or transmit the corresponding token to the POS terminal. In other embodiments, the vehicle head unit **14** may transmit the token to the electronic circuitry **50** each time the electronic circuitry **50** is called upon in the payment process. The electronic circuitry **50** is described in more detail below with reference to FIG. 2B.

C. Exemplary POS Terminal

(36) The POS terminal **20** may be a combination of hardware and software components and may include a display, a camera, and/or one or more processor(s) such as a microprocessor coupled to a memory. The POS terminal **20** may also include a wired and/or wireless transceiver for transmitting/receiving radio signals using NFC, Bluetooth, DSRC, RFID, USB, Wi-Fi, etc. The POS terminal **20** may communicate with the

vehicle head unit **14** and/or the electronic circuitry via a short-range and/or very short-range communication link. Additionally, the POS terminal **20** may communicate with other devices via a wired and/or wireless long-range communication network **130** such as a proprietary network, a secure public Internet, a wide area network (WAN), the Internet, and/or some other type of network, such as dedicated access lines, plain ordinary telephone lines, satellite links, combinations of these, etc. Where the digital network **130** comprises the Internet, data communication may take place over the digital network **130** via an Internet communication protocol. For example, upon receiving a token and/or financial data for a financial card the POS terminal **20** may transmit the received data to an acquiring bank server which may in turn transmit the received data to an issuing bank server for the financial card. The issuing bank server may then transmit an indication back to the POS terminal **20** that the payment has been accepted. For example, the issuing bank server may communicate with the third party token server **104** to retrieve the financial data corresponding to the token and generate a charge for the corresponding financial card. Alternatively, the issuing bank server may obtain the financial card corresponding to the token and/or financial data in any other suitable manner. In other scenarios, when the received data cannot be verified at the issuing bank server or the corresponding financial card does not have a sufficient balance and/or sufficient credit to make the payment, the issuing bank server may transmit an indication to the POS terminal **20** that the payment has been declined.

(37) While the POS terminal **20** is depicted in FIG. **1** as attached to a fuel station, this is merely one example embodiment. The POS terminal **20** may be a stand-alone device, may be attached to a drive-thru window at a drive-thru and/or car wash, and/or may be presented in any other suitable manner.

D. Exemplary Token Server

(38) According to embodiments, the third-party token server **104** may be a combination of hardware and software components, also as described in more detail below. The third-party token server **104** may have an associated database for storing tokens, the financial data represented by the tokens, and/or the financial cards corresponding to the financial data. Moreover, the third-party token server **104** may include one or more processor(s) such as a microprocessor coupled to a memory.

(39) The memory may be tangible, non-transitory memory and may include any types of suitable memory modules, including random access memory (RAM), read-only memory (ROM), flash memory, other types of persistent memory, etc. The memory may store, for example instructions executable on the processors for generating/transmitting tokens and/or storing the tokens with associated financial data/financial cards.

(40) It will be appreciated that although only one third-party token server **104** is depicted in FIG. **1**, multiple third-party token servers **104** may be provided for the purpose of distributing server load, serving different web pages, etc. These multiple third-party token servers **104** may include a web server, an entity-specific server (e.g. an Apple® server, etc.), a server that is disposed in a retail or proprietary network, etc.

(41) The third-party token server **104** may communicate with the vehicle head unit **14** via the network **130**. For example, the vehicle head unit **14** may transmit financial data for a financial card to the third-party token server **104** and in turn, the third-party token server may generate/transmit a token representing the financial data to the vehicle head unit **14**. The vehicle payment application in the vehicle head unit **14** may then store the token with an indication of the financial card and/or transmit the token when making a payment with the corresponding financial card.

II. Exemplary System Hardware

A. Exemplary Vehicle Head Unit

(42) Turning now to FIG. **2A**, the vehicle head unit **14**, may include a display **236** similar to the display **18** as shown in FIG. **1**, a communication unit **238**, a user-input device (not shown), a camera and/or other image sensor (not shown), and/or a controller **224**. The controller **224** may include a program memory **226**, a microcontroller or a microprocessor (MP) **228**, a random-access memory (RAM) **230**, and/or an input/output (I/O) circuit **234**, all of which may be interconnected via an address/data bus **232**. The program memory **226** and the microprocessor **228** may be similar to the memory and processor respectively, as described in FIG. **1**.

(43) The program memory **226** may include an operating system **260**, a data storage **262**, a plurality of software applications **264**, and/or a plurality of software routines **268**. The operating system **260**, for example, may include iOS®, Android™, Palm® webOS, Windows Mobile/Phone, BlackBerry® OS, Symbian® OS, Microsoft Windows®, OS X®, Linux®, Unix®, etc. The data storage **262** may include data such as user profiles, indications of financial cards, tokens, application data for the plurality of applications **264**, routine data for the plurality of routines **268**, and/or other data necessary to interact with the third-party token server **104**, electronic circuitry **50**, and/or POS terminal **20** through the digital network **130**, and/or short-range/very short-range communication links **120**, **122**. In some embodiments, the controller **224** may

also include, or otherwise connected to, other data storage mechanisms (e.g., one or more hard disk drives, optical storage drives, solid state storage devices, etc.) that reside within the vehicle head unit **14**.

(44) The communication unit **238** may communicate with the third-party token server **104** via any suitable wireless communication protocol network, such as a wireless telephony network (e.g., GSM, CDMA, LTE, etc.), a Wi-Fi network (802.11 standards), a WiMAX network, etc. Additionally, the communication unit **238** may include one or more transceivers and communicate with the electronic circuitry **50** and/or POS terminal via any suitable short-range/very short-range wired or wireless communication protocol network, such as USB, Bluetooth, Wi-Fi, NFC, RFID, DSRC, etc. The user-input device (not shown) may include a “soft” keyboard that is displayed on the display **236** of the vehicle head unit **14**, an external hardware keyboard such as the hardware input controls **22** as shown in FIG. **1**, or any other suitable user-input device.

(45) It should be appreciated that although FIG. **2A** depicts only one microprocessor **228**, the controller **224** may include multiple microprocessors **228**. Similarly, the memory of the controller **224** may include multiple RAMs **230** and/or multiple program memories **226**. Although FIG. **2A** depicts the I/O circuit **234** as a single block, the I/O circuit **234** may include a number of different types of I/O circuits. The controller **224** may implement the RAM(s) **230** and/or the program memories **226** as semiconductor memories, magnetically readable memories, and/or optically readable memories, for example.

(46) The one or more processors **228** may be adapted and configured to execute any one or more of the plurality of software applications **264** and/or any one or more of the plurality of software routines **268** residing in the program memory **226**, in addition to other software applications. One of the plurality of applications **264** may be a vehicle payment application **266** that may be implemented as a series of machine-readable instructions for performing the various tasks associated with receiving information at, displaying information on, and/or transmitting information from the vehicle head unit **14**.

(47) One of the plurality of applications **264** may be a native application and/or web browser **270**, such as Apple's Safari®, Google Chrome™, Microsoft Internet Explorer®, and Mozilla Firefox® that may be implemented as a series of machine-readable instructions for receiving, interpreting, and/or displaying web page information while also receiving inputs from the user. Another application of the plurality of applications may include an embedded web browser **276** that may be implemented as a series of machine-readable instructions for receiving, interpreting, and/or displaying web page information. One of the plurality of routines may include a financial card storage routine **272** which obtains financial data for a financial card from the user and/or a portable device, transmits the financial data to a third-party token server **104** and stores a received token with an indication of the financial data at the vehicle payment application **266**. Another routine in the plurality of routines may include a payment routine **274** that receives a selection of a stored financial card and transmits a token representing financial data for the financial card to the POS terminal **20** for making a payment.

(48) Preferably, a user may launch the vehicle payment application **266** from the vehicle head unit **14** to communicate with the third-party token server **104**, the electronic circuitry **50** and/or the POS terminal **20** to implement the vehicle payment system **100**. Additionally, the user may also launch or instantiate any other suitable user interface application (e.g., the native application or web browser **270**, or any other one of the plurality of software applications **264**) to access the third-party token server **104**, the electronic circuitry **50** and/or the POS terminal **20** to realize the vehicle payment system **100**.

(49) In one embodiment, to interact with the vehicle payment system **100**, the user may execute the vehicle payment application **266** on the vehicle head unit **14**. Using the vehicle payment application **266**, the user may make payments by navigating a series of vehicle payment application screens. FIGS. **3A-E** depict vehicle payment application pages and/or screens that may be displayed on the vehicle head unit **14** in various embodiments of the vehicle payment system **100**. However, the screens depicted in FIGS. **3A-E** are merely illustrations of an exemplary embodiment. In some embodiments, the vehicle head unit **14** may display web pages. While the vehicle payment application **266** may retrieve application data such as indications of financial cards, tokens, user profiles, etc. from the data storage **262** as described above, the application data may also be stored in one or more server devices (not shown). In some embodiments, the vehicle payment application **266** may communicate with the one or more server devices to transmit/receive server data and/or to implement the vehicle payment system **100**.

(50) In any event, the user may launch the vehicle payment application **266** from the vehicle head unit **14** via any suitable manner, such as touch-selecting a vehicle payment application icon (not shown) on the display **236** of the vehicle head unit **14** and/or using one of the hardware input controls **22**. After the user launches

the vehicle payment application 266, a payment activation screen of the vehicle payment application 266 may be displayed to the user on the vehicle head unit.

B. Exemplary Electronic Circuitry

(51) Referring now to FIG. 2B, the electronic circuitry 50 (also referred to herein as an “electronic device”) may include an NFC module 240 and a communication unit 258. The electronic circuitry 50 may be proximately attached to the exterior of the vehicle 12. Locations on the interior of the vehicle 12 which are within several inches or a foot from the exterior of the vehicle 12 may be proximate to the exterior of the vehicle 12. For example, the electronic circuitry 50 may be attached to a door panel or fuel cap of the vehicle 12 on the exterior of the vehicle 12. In another example, the electronic circuitry 50 may be attached to a fuel cap from the interior of the vehicle 12 and proximate to the exterior of the vehicle 12. In this manner, the electronic circuitry 50 may be placed in very close proximity of a POS terminal 20 (e.g., within one inch, three inches, six inches, a foot, three feet, etc.) to securely transmit financial data or a token representing the financial data.

(52) In some embodiments, like the vehicle head unit 14, the electronic circuitry 50 may also include a controller 242. Similar to the controller 224, the controller 242 may include a program memory 246, a microcontroller or a microprocessor (MP) 248, a random-access memory (RAM) 250, and/or an input/output (I/O) circuit 254, all of which may be interconnected via an address/data bus 252. The program memory 246 may include a data storage and/or a plurality of software routines. The data storage may include data such as indications of financial cards, tokens, etc. In some embodiments, the controller 242 may also include, or otherwise be communicatively connected to, other data storage mechanisms (e.g., one or more hard disk drives, optical storage drives, solid state storage devices, etc.) that reside within the electronic circuitry 50.

(53) The communication unit 258 may include one or more transceivers, and communicate with the vehicle head unit 14 and/or the POS terminal 20 via any suitable short-range/very short-range wired or wireless communication protocol network, such as USB, Bluetooth, Wi-Fi, NFC, RFID, DSRC, etc. For example, using the NFC module 240, the communication unit 258 may communicate with the POS terminal using NFC.

(54) As discussed with reference to the controller 224, it should be appreciated that although FIG. 2B depicts only one microprocessor 248, the controller 242 may include multiple microprocessors 248. Similarly, the memory of the controller 242 may include multiple RAMs 250 and/or multiple program memories 246. Although the FIG. 2B depicts the I/O circuit 254 as a single block, the I/O circuit 254 may include a number of different types of I/O circuits. The controller 242 may implement the RAM(s) 250 and/or the program memories 246 as semiconductor memories, magnetically readable memories, and/or optically readable memories, for example.

(55) The one or more processors 248 may be adapted and configured to execute any one or more of a plurality of software routines residing in the program memory 246. One of the plurality of routines may include a device recognition routine which identifies devices within range for communication, such as a POS terminal 20 which may communicate with the electronic circuitry 50. Another routine in the plurality of routines may include a token transmission routine which retrieves a token from data storage in response to receiving an indication of a selected financial card from the vehicle head unit 14. The token may also be retrieved from the vehicle head unit 14. The token transmission routine may then transmit the obtained token corresponding to the financial card to a POS terminal.

III. Exemplary Screenshots of a Vehicle Payment Application

(56) In the present aspects, the following screenshots shown in FIGS. 3A-E are examples of what may be displayed to a user as part of a vehicle payment application 266. In the present aspects, the following exemplary screenshots shown in FIGS. 3A-E may be displayed on the vehicle head unit 14 as shown in FIG. 1. As will be appreciated by those of ordinary skill in the relevant art(s), the exemplary screenshots shown in FIGS. 3A-E are for illustrative purposes, and their associated functionality may be implemented using any suitable format and/or design for facilitating their corresponding described functionalities without departing from the spirit and scope of the present disclosure.

(57) Moreover, the following exemplary screenshots shown in FIGS. 3A-E may be presented on the display 236 of the vehicle head unit 14 when the vehicle is operated by a user. When the vehicle 12 is an autonomous vehicle, the autonomous vehicle may receive and/or interact with the information presented by the vehicle payment application 266 without presenting all of the exemplary screenshots. For example, the autonomous vehicle may receive indications of stored financial cards and/or select one of the stored financial cards for making a payment without the indications being presented on the display 236 of the vehicle head

unit **14**. In other embodiments, the exemplary screenshots may be presented on the vehicle head unit **14** of the autonomous vehicle to allow users within the autonomous vehicle to interact with the vehicle payment application **266**.

A. Exemplary Financial Card Entry Screen

(58) To add a new financial card or financial account (also referred to herein as a “virtual account”) to the vehicle payment application **266**, a user may select a user control such as an “Add Credit/Debit Card” button on a home screen (not shown) and/or a financial card selection screen of the vehicle payment application **266**. With reference now to FIG. **3A**, a financial card entry screen **300** may be displayed on the vehicle head unit **14** when the user selects the “Add Credit/Debit Card” button on the home screen, financial card selection screen, and/or any other suitable screen of the vehicle payment application **266**.

(59) The financial card entry screen **300** may include a user control **304** for receiving an image of the financial card at the vehicle head unit **14**. For example, the vehicle payment application **266** may instruct the camera within the vehicle head unit **14** to capture an image of the user's financial card. The financial card entry screen **300** may include an instruction to position the financial card within a frame **302**, where the frame is included in the user control **304**. When a user positions the boundaries of a financial card within the frame of the user control **304**, the camera of the vehicle head unit **14** may automatically capture an image of the financial card. In other embodiments, the user may select a user control instructing the capture to capture an image of the financial card when the boundaries of the financial card are positioned within the frame. The camera may capture one and/or both sides of the financial card.

(60) Additionally, the financial card entry screen **300** may include user controls **306-312** for manually entering financial data for the financial card, such as a credit/debit card number **306**, a cardholder name **308**, an expiration date **310**, a card type **312** (e.g., VISA®, MasterCard®, American Express®, Discover®, etc.), a CSC code, and/or any other suitable information such as a billing address, a card nickname, etc.

(61) When an image of the financial card and/or the financial data for the financial card is provided at the financial card entry screen **300**, the user may select a “Submit” button **314** to store an indication of the financial card at the vehicle payment application **266**. The vehicle payment application **266** may use various image processing techniques, such as optical character recognition (OCR) to identify the financial data included in the image of the financial card. Additionally, the indication of the financial card may be an edited version of the image of the financial card. At least some of the financial data, such as the cardholder name, card number, expiration date and/or CSC code, may be removed and/or masked using various image processing techniques. Accordingly, the edited version of the image of the financial card may display the shape of the financial card, the background color/image of the financial card, the name of the issuer of the financial card (e.g., Chase™, Bank of America™, Wells Fargo™ Citigroup™, PNC™, Capital One™, etc.), the card type (e.g., VISA®, MasterCard®, American Express®, Discover®, etc.), masked financial data such as the last four digits of the financial card, etc. When an image of the financial card is not captured and the user manually enters the financial data for the financial card instead, the vehicle payment application **266** may create an image of the financial card based upon the entered card type. For example, if the card type is American Express®, the vehicle payment application **266** may generate an image of a financial card having a green background color, and/or the name “American Express.”

(62) In some embodiments, when the “Submit” button **314** is selected, the vehicle payment application **266** may transmit the financial data retrieved from the image of the financial card **304** or manually entered via the user controls **306-310**, to a third-party token server **104** as shown in FIG. **1**. The third-party token server **104** may then generate a token which may be a string of randomly generated alphanumeric or numeric characters that represents the financial data. For example, the token may be a 16 digit alphanumeric character string. The third-party token server **104** may store the token with the financial data that the token represents and/or may transmit the token to the vehicle payment application **266** for additional storage. In some embodiments, the third-party token server **104** may provide the financial data to the issuing bank for the financial card after receiving the token from the issuing bank when a payment is processed. In other embodiments, the third-party token server **104** may be the issuing bank. The tokenization process may be completed in any number of ways including but not limited to the system and method for secure acceptance of customer credit card numbers as described in U.S. patent application Ser. No. 14/505,599, which is incorporated by reference herein.

(63) In any event, the vehicle payment application **266** may then store the indication of the financial card so that the user may select the financial card for making a payment at a financial card selection screen, for example. In some embodiments, the token representing financial data for the financial card may be stored

with the indication of the financial card. The token may be transmitted when the user selects the financial card for making the payment. In other embodiments, the vehicle payment application **266** may transmit the token to the electronic circuitry **50** as shown in FIG. **1**, and/or the electronic circuitry **50** may store the token. In such embodiments, when a user selects a financial card, the vehicle payment application **266** may transmit an indication to the electronic circuitry **50** that the financial card has been selected, and the electronic circuitry may retrieve and/or transmit the token corresponding to the selected financial card.

(64) In addition to capturing an image of the financial card and/or manually entering financial data for the financial card at the financial card entry screen **300**, the financial card entry screen **300** may include a user control (not shown) to import the financial card from another application such as a photo library, email application, etc., and/or an external source such as a portable device **10** in communication with the vehicle head unit **14**. In this manner, the user may import indications of financial cards and/or financial data for the financial card from another application and/or from the portable device **10** to the vehicle payment application **266**.

B. Exemplary Activation Screen

(65) When the vehicle detects a POS terminal **20**, the vehicle payment application **266** may display a payment activation screen **320** as depicted in FIG. **3B**. For example, the vehicle head unit **14** and/or electronic circuitry **50** proximately attached to the exterior of the vehicle may detect the presence of a Bluetooth signal, RFID signal, NFC signal, DSRC signal, Wi-Fi signal, etc. In another example, the vehicle head unit **14** and/or electronic circuitry **50** proximately attached to the exterior of the vehicle may determine that the vehicle is within a predetermined distance of a POS terminal **20**. More specifically, the vehicle head unit **14** may obtain locations of merchants (e.g., fuel stations, dry cleaners, fast food or other restaurants, coffee shops, grocery stores, pharmacies, vehicle repair shops, etc.) within a geographic area surrounding the current location of the vehicle head unit **14**. The vehicle head unit **14** and/or electronic circuitry **50** proximately attached to the exterior of the vehicle may determine that the vehicle is within a predetermined distance of a POS terminal **20** associated with one of the merchants, when the current location of the vehicle is within the predetermined distance (e.g., 50 feet, 100 feet, 200 feet, etc.) of one of the merchants.

(66) In yet another example, a camera located within the vehicle head unit **14** or other portion of the vehicle may capture images of an area surrounding the vehicle. The images may be visible light images, radar images, infrared images, or any other suitable image within the electromagnetic spectrum including the visible and invisible light spectrums. The vehicle head unit **14** may then analyze these images using various digital image processing techniques (such as object recognition, optical character recognition etc.), to identify merchants as well as their respective locations relative to the location of the vehicle. When a merchant is identified that is within a predetermined distance of the vehicle, the vehicle may detect a POS terminal **20**.

(67) In some embodiments, the vehicle head unit **14** and/or electronic circuitry **50** may also identify the device transmitting a signal and/or a name of a corresponding network. When the device transmitting the signal and/or the name of the corresponding network is associated with a POS terminal **20**, the vehicle payment application **266** may detect a POS terminal **20**. In other embodiments, the vehicle payment application **266** may detect a POS terminal **20** upon detecting the presence of one of the signals mentioned above. Additionally, the vehicle head unit **14** may obtain a list of POS terminals which accepts payments from a vehicle and/or any other type of mobile payments. For example, the vehicle head unit **14** may communicate with a third party server, such as a Department of Motor Vehicles (DMV) server to obtain the list. The detected POS terminal **20** may be compared to the list of POS terminals to determine whether the detected POS terminal **20** accepts vehicle payments. Furthermore, the user may provide a list of preferred merchants and/or merchant locations. The merchant and/or merchant location associated with the detected POS terminal **20** may be compared to the list of preferred merchants to determine whether the user authorizes vehicle payments at the detected POS terminal **20**. If the user does not authorize vehicle payments or the detected POS terminal **20** does not accept vehicle payments, the user may manually purchase goods or services provided by the merchant or may navigate to another merchant location.

(68) In any event, upon detecting the POS terminal **20**, the vehicle payment application **266** may automatically activate and/or display the payment activation screen **320**. In other embodiments, the user may activate the vehicle payment application **266** by touch-selecting a vehicle payment application icon on the display **236** of the vehicle head unit **14**, as described above. In yet other embodiments, the POS terminal **20** may identify a vehicle. For example, the POS terminal **20** may detect the presence of a Bluetooth signal, RFID signal, NFC signal, DSRC signal, Wi-Fi signal, etc. In another example, the POS terminal may

capture, with a camera communicatively coupled to the POS terminal **20**, images of an area surrounding the POS terminal **20**. The POS terminal **20** may then analyze these images using various digital image processing techniques (such as object recognition, optical character recognition etc.), to identify a vehicle and/or a vehicle license plate number. In yet another example, the POS terminal **20** may receive an indication from the vehicle head unit **14** and/or electronic circuitry **50** that the vehicle **12** is within a predetermined range of the POS terminal **20**. As a result, the POS terminal **20** may transmit the payment activation screen **320** to the vehicle head unit **14** and/or electronic circuitry **50**. In addition to displaying the payment activation screen **320**, the POS terminal **20**, the vehicle head unit **14**, and/or the electronic circuitry **50** may establish a communication link between the POS terminal **20** and the vehicle head unit **14** and/or the electronic circuitry **50**. The communication link may be a short-range communication link (e.g., Bluetooth, DSRC, RFID, Wi-Fi, USB, etc.), a very short-range communication link (e.g., NFC, HF RFID, etc.) or any suitable combination.

(69) In any event, the payment activation screen **320** may include a prompt **322** asking the user whether she would like to pay from her vehicle, a “YES” button **324**, and/or a “NO” button **326**. If the user selects the “YES” button **324** by for example, touch-selecting the display **236** of the vehicle head unit **14** or using one of the hardware input controls, the vehicle payment application **266** may present a financial card selection screen on the vehicle head unit **14** for selecting one of the stored financial cards, as described in more detail below. On the other hand, if the user selects the “NO” button **326**, the vehicle payment application **266** may automatically close.

(70) In some embodiments, before presenting the financial card selection screen, the POS terminal **20** may transmit data over a short-range communication link **120** to the vehicle head unit **14**, and/or over first and second short-range/very short-range communication links **120**, **122** to the electronic circuitry **50** and then to the head unit **14**. For example, when the establishment associated with the POS terminal **20** is a fuel station, the data may include a request for the user to provide a financial card to the POS terminal **20** before filling up with fuel, which may be presented on the display **236** of the vehicle head unit **14**. Then when the fueling process is over, the POS terminal **20** may transmit additional data including the amount of fuel provided to the vehicle, the total cost of the fuel, and/or a request to authorize payment for the total cost of the fuel using the financial card provided to the POS terminal **20**. The additional data may be presented on the display **236** of the vehicle head unit **14**.

(71) In another example, the establishment associated with the POS terminal **20** may be another type of establishment which provides goods or services at or nearby a user's vehicle, such as a drive-thru or a car wash. The POS terminal **20** may transmit data including the total cost of the goods or services and/or a request for the user to transmit payment to the POS terminal **20**. In some embodiments, the user may be prompted to provide permission for the vehicle head unit **14** and/or the electronic circuitry **50** to receive the data from the POS terminal **20**.

(72) In some embodiments, the vehicle payment application **266** may store user preferences and/or other information indicative of retailers/merchants preferred by the user. For example, the user may provide indications of preferred retailers/merchants and/or services, goods, or items that the user likes to purchase via the display **236** of the vehicle head unit **14**. Additionally, the vehicle payment application **266** may identify user preferences based upon previous purchases made by the user and the establishments and/or types of services, goods, or items that the user typically purchases.

(73) When the vehicle head unit **14** and/or the electronic circuitry **50** detects the presence of short-range/very short-range wireless signal and/or identifies the device transmitting the signal, the vehicle payment application **266** may determine whether the device is associated with an establishment (also referred to herein as a “point of interest (POI)”) and/or services, goods, or items which are preferred by the user. When the device is associated with a preferred establishment/item and is within a threshold distance of the vehicle **12**, the vehicle payment application **266** may present a message to the user that a preferred establishment/item is nearby. The vehicle payment application **266** may also present an indication of the distance to the preferred establishment/item and/or an indication of the preferred establishment/item such as a name of the preferred establishment/item. Moreover, the vehicle payment application **266** may also display a map of a geographic area including the user's current location with an indication, such as a pin which marks the location of the preferred establishment/item. In another embodiment, the map display may include several indications which mark the locations of preferred establishments/items and/or any other merchants which accept vehicle payments. The vehicle payment application **266** may also determine the distance between the vehicle and each of the preferred establishments/items and/or other merchants based upon their

respective positions within the map display.

(74) In some embodiments, the vehicle payment application **266** may provide navigation directions to the location of the preferred establishment/item or a location of one of several preferred establishments/items that is selected by a user. For example, the user may touch-select one of the preferred establishment/items on the map display. In another example, the user may provide a type of goods or services that the user wants to purchase. For example, the user may request a fast food restaurant. The vehicle payment application **266** may then identify one of the preferred establishment/items included in the map display that provides the type of goods or services that the user wants to purchase. In yet another example, when the vehicle is an autonomous vehicle, the vehicle payment application **266** may identify a type of goods or services that needs to be purchased, such as fuel for the vehicle. The vehicle payment application **266** may then identify one of the preferred establishment/items included in the map display that provides fuel. For example, the vehicle payment application **266** may identify the closest preferred merchant location that provides fuel and/or any other suitable merchant location on the map display that provides fuel.

(75) The navigation directions may be a set of turn-by-turn navigation instructions for traversing from the current location of the vehicle to the location of the preferred establishment/item. In some embodiments, the vehicle payment application **266** may communicate with a third party server, such as a map server and/or a navigation server which may provide the map data and/or navigation data for displaying a map and navigation directions. For example, the map server may provide indications of several merchant locations to the vehicle payment application **266**. The vehicle payment application **266** may then determine which of the merchant locations are within a geographic area surrounding the user's current location. Accordingly, the vehicle payment application **266** may generate a map display of the geographic area and include the indications of merchant location within the geographic area on the map display. The map display may be presented on display **18** of the vehicle head unit **14**.

(76) When the vehicle is driven by the user, the navigation directions may be provided audibly via the speakers of the vehicle head unit **14** or visually, where the navigation directions overlay the map display. For example, the vehicle payment application **266** may highlight or otherwise annotate the roads, streets, etc. included in the turn-by-turn navigation directions for traversing from the current location of the vehicle to the location of the preferred establishment/item. When the vehicle is an autonomous vehicle, the navigation directions may be a set of instructions provided to the autonomous vehicle for navigating to the preferred establishment/item. The instructions may include indications of upcoming maneuvers such as turns, distances in which an upcoming maneuver needs to be made (e.g., turning left in three miles), and/or any other suitable instructions for guiding the autonomous vehicle to the preferred establishment/item.

(77) Also in some embodiments, when the user selects the "YES" button **324** indicating she would like to make a payment from her vehicle, the vehicle payment application **266** may authenticate the user to ensure she is authorized to use the stored financial cards to make payments. The vehicle payment application **266** may store biometric information for authorized users, such as fingerprint data, facial data, retinal data, voice data, weight data, etc. To authenticate the user, the vehicle payment application **266** may retrieve a biometric identifier from the user. For example, the vehicle head unit **14** may capture an image of the user, a microphone in the vehicle head unit **14** may capture voice data for the user, the user may press a finger on the vehicle head unit **14** for the vehicle payment application **266** to receive fingerprint data, weight or pressure sensors in the vehicle may determine the weight of the user, and/or a biometric identifier for the user may be obtained in any other suitable manner. The vehicle payment application **266** may then verify the received biometric identifier by comparing the obtained biometric identifier to stored biometric information for authorized users. If the obtained biometric identifier matches the stored biometric information, the vehicle payment application **266** may determine the user is an authorized user and present the financial card selection screen, as described below.

(78) In some scenarios, for example when the vehicle is an autonomous vehicle, the vehicle payment application **266** may authenticate the vehicle to ensure that the vehicle is authorized to use the stored financial cards to make payments. For example, a camera within the vehicle head unit **14** or another portion of the vehicle may capture an image of the license plate. Using digital image processing techniques such as objection recognition and/or optical character recognition, the vehicle head unit **14** may identify the license plate number for the vehicle. The license plate number may be compared to a stored license plate number for an authorized vehicle. If the obtained license plate number matches the stored license plate number, the vehicle payment application **266** may determine the vehicle is an authorized vehicle and present the financial card selection screen, as described below. In some embodiments, the vehicle payment application **266** and/or

the POS terminal **20** may transmit the license plate number or other vehicle identification information (e.g., a make/model of the vehicle, a vehicle identification number (VIN), etc.) to a third party server for verifying that the autonomous vehicle has not been stolen. The third party server may then compare the vehicle identification information to stolen vehicle records to ensure that the autonomous vehicle has not been stolen.

(79) In other embodiments, the POS terminal **20** may authenticate the vehicle and/or a user within the vehicle by receiving a biometric identifier from the user and/or vehicle identification information. The biometric identifier and/or vehicle identification information may be compared to a database of authorized users or vehicles, and if there is a match, the POS terminal **20** may verify that the user and/or vehicle is authorized to make vehicle payments. For example, when an autonomous vehicle is authorized to make vehicle payments using a particular financial card, a user may register the autonomous vehicle as an authorized vehicle. In some embodiments, the financial card or virtual account may also be registered with the user and/or vehicle. Accordingly, when the user and/or vehicle transmits financial data/virtual account information for the financial card/virtual account to the POS terminal **20**, the POS terminal **20** may verify that the user and/or vehicle is authorized to make payments with the financial card/virtual account.

(80) The vehicle payment application **266** may also store a username and/or password, a PIN number, and/or any other unique identifier for authorized users. To authenticate the user, the vehicle payment application **266** may receive login information, such a username and password, a PIN number, or identification information from an authenticated key fob and verify the login information by comparing the login information to the stored unique identifier. When there is a match, the vehicle payment application **266** may determine that the user is an authorized user. In yet other embodiments, the vehicle head unit **14** may detect the presence of a portable device **10**, for example from a short-range/very short-range communication link. The portable device **10** may be used to authenticate the user by transmitting a unique identifier to the vehicle head unit **14** and/or transmitting any other suitable information indicating that the portable device **10** belongs to an authorized user. If the user is not authenticated, the vehicle payment application **266** may not display the financial card selection screen and/or may continue to receive a biometric identifier and/or login information for the user.

(81) As mentioned above, when the vehicle **12** is an autonomous vehicle, the vehicle payment application **266** may not display the payment activation screen **320**. Instead, the autonomous vehicle may communicate with the vehicle payment application **266** to initiate the payment process.

C. Exemplary Financial Card Selection Screen

(82) In response to the user selecting the “YES” button **324** on the payment activation screen **320** and/or authenticating the user, the vehicle payment application **266** may present a financial card selection screen **340** as depicted in FIG. 3C. The vehicle payment application **266** may also present the financial card selection screen **340** in response to a request from the POS terminal **20** to provide a financial card and/or transmit payment to the POS terminal. In any event, the financial card selection screen **340** may include indications of financial cards **342-348** which were added at the financial card entry screen **300** as shown in FIG. 3A. The financial card selection screen **340** may also include user controls for selecting the financial cards. For example, a user may select the Discover® card **346** by touch-selecting the area of the display **236** which includes the Discover® card **346**.

(83) As mentioned above, each of the indications of financial cards **342-348** may include an edited version of the image of the financial card where at least some of the financial data, such as the cardholder name, card number, and/or expiration date, is removed and/or masked. Accordingly, the edited version of the image of the financial card may display the shape of the financial card, the background color/image of the financial card, the name of the issuer of the financial card (e.g., Chase™, Bank of America™, Wells Fargo™, Citigroup™, PNC™, Capital One™, etc.), the card type (e.g., VISA®, MasterCard®, American Express®, Discover®, etc.), masked financial data such as the last four digits of the card number, etc. For example, the indication of the VISA® card **344** includes a rectangular image having Bank of America™ as the name of the issuing bank, VISA® as the card type, and a red background color. The user may touch-select the VISA® card **344**, and as a result the vehicle payment application **266** may present a payment authorization screen, as described in more detail below in FIG. 3D.

(84) While the indications of financial cards **342-348** are displayed adjacent to each other in the financial card selection screen **340**, this is merely an exemplary embodiment. In other embodiments, the indications of financial cards **342** may be displayed such that they are layered on top of each other where a portion of some of the indications **342-348** is occluded. Moreover, the indications of financial cards **342-348** may be displayed

on the financial card selection screen **340** in any suitable manner to allow the user to select one of the financial cards. Furthermore, additional and/or alternative information may be included in the indications of financial cards, such as a nickname for the financial card, a billing address for the financial card, and/or any other suitable information.

(85) As mentioned above, when the vehicle **12** is an autonomous vehicle, the vehicle payment application **266** may not display the financial card selection screen **340**. Instead, the autonomous vehicle may communicate with the vehicle payment application **266** to select a financial card. For example, the autonomous vehicle may select one of the stored financial cards based upon the credit limits for the stored financial cards, an amount of money corresponding to each of the stored financial cards, a preference for a particular financial card previously indicated by a user, or in any other suitable manner.

D. Exemplary Payment Authorization Screen

(86) Turning now to FIG. **3D**, a payment authorization screen **360** may be presented by the vehicle payment application **266**, when the user selects a financial card for making a payment. The payment authorization screen **360** may include an indication of the selected financial card **362**, a request to transmit payment **364**, and/or “YES” and “NO” buttons **366**, **368**. In some embodiments, the payment authorization screen **360** may also include an indication of the goods or services purchased and/or a total cost of the goods or services provided by the establishment associated with the POS terminal **20**. When the establishment is a fuel station, the total cost may not be available and instead, the user may provide the financial card to the POS terminal **20** before receiving fuel. Then when the fueling process is finished, the vehicle payment application **266** may present an additional screen (not shown) to authorize payment for the total cost of the fuel. The additional screen may display the total cost.

(87) In some embodiments, the POS terminal **20** may also transmit links to loyalty reward programs, which may be presented on the payment authorization screen **360**. In this manner, the user may receive loyalty reward points for transmitting the payment and/or may receive discounts on the purchase. Further, the vehicle payment application **266** may include spending limits for each of the stored financial cards, which may be monthly spending limits, yearly spending limits, etc. For example, a parent and/or guardian of the user may set a monthly spending limit on one of the stored financial cards. When transmitting the payment would cause the user to exceed the allotted spending limit for the financial card, the payment authorization screen **360** may include a message indicating that the payment would cause the financial card to exceed the spending limit and/or requesting the user to select another financial card. In some scenarios, the user may be able to override the spending limit, for example by entering an emergency override code, receiving permission from the parent and/or guardian who entered the spending limit, etc.

(88) In any event, when the user selects the “YES” button **366**, the POS terminal **20** may receive a token representing financial data (e.g., a cardholder name, a card number, an expiration date, card type, and/or a CSC code) for the financial card. For example, the vehicle payment application **266** may transmit the token for the financial card to the POS terminal **20** via a short-range communication link. In another example, the vehicle payment application **266** may transmit the token for the financial card to the electronic circuitry **50** proximately attached to the exterior of the vehicle **12** via a first short-range communication link, which may in turn transmit the token for the financial card to the POS terminal **20** via a second very short-range communication link, such as NFC, HF RFID, etc.

(89) In yet another example, the vehicle payment application **266** may have transmitted the token for the financial card to the electronic circuitry **50** when the financial card was initially added to the vehicle payment application **266**. When the user selects the financial card by selecting the “YES” button **366**, the vehicle payment application **266** may transmit an indication of the selected financial card to the electronic circuitry **50**. The indication may be the card type, the name of the issuing bank, the last four digits of the card number, and/or a combination of these. The electronic circuitry **50** may in turn, retrieve the token corresponding to the selected financial card, and/or transmit the token to the POS terminal **20**. The POS terminal **20** may then use the token to process the payment from the selected financial card.

(90) On the other hand, when the user selects the “NO” button **368**, the vehicle payment application **266** may display the financial card selection screen **340** as shown in FIG. **3C** to allow the user to select another financial card.

(91) When the establishment associated with the POS terminal **20** is a fuel station, the vehicle payment system **100** may include further functionality in addition to making payments from a vehicle. For example, when the user selects the “YES” button **366** to transmit the token for the financial card to the POS terminal **20**, the vehicle payment application **266** may cause the fuel cap of the vehicle **12** to automatically be opened

for receiving fuel. In some embodiments, the fuel cap of the vehicle **12** may be a door which may be attached to a door opening/closing mechanism such as a lever. The electronic circuitry **50** proximately attached to the exterior of the vehicle **12** may be communicatively coupled to the door opening/closing mechanism and may control the door opening/closing mechanism. In addition to transmitting the token for the financial card, the vehicle payment application **266** may transmit a request to the electronic circuitry **50** to open the door. In other embodiments, the vehicle head unit **14** may be communicatively coupled to the door opening/closing mechanism and may transmit a control signal to the door opening/closing mechanism to open the door.

(92) The fuel tank of the vehicle **12** may also open automatically when the nozzle which provides the fuel makes contact with the fuel tank opening. For example, the fuel tank opening may expand when receiving pressure from the nozzle and/or condense when the nozzle is removed from the fuel tank opening. In some embodiments, the fuel tank opening may be made of an elastic material which stretches when receiving pressure from the nozzle.

(93) Additionally in some embodiments, the fuel pump at the fuel station may be a smart fuel pump and the nozzle may be controlled automatically, for example via a robotic/mechanical arm. Accordingly, when the POS terminal **20** receives the token and the fuel cap is opened, the vehicle head unit **14** and/or the electronic circuitry **50** may transmit a signal to the POS terminal **20** indicating that the fuel cap is opened and requesting the POS terminal **20** to provide fuel to the vehicle **12**. In some embodiments, the robotic/mechanical arm may include one or several sensors to detect the height and/or position of the fuel tank opening (e.g., via proximity sensors, such as ultrasonic, infrared, or radar sensors or any other suitable sensors that transmit light beams to detect distance and/or shape) and/or to detect that the fuel tank is open (e.g., via various image processing techniques). The robotic/mechanical arm and/or the smart fuel pump may also calculate the distance between the fuel pump and the fuel tank opening and/or compare this distance to a predetermined maximum distance for placing the nozzle within the fuel tank opening. In other embodiments, the vehicle head unit **14** and/or the electronic circuitry **50** may transmit the position of the fuel tank opening to the POS terminal **20**, including the height off the ground of the fuel tank opening, the distance between the fuel tank opening and the POS terminal, etc. When the fuel tank opening is within the predetermined maximum distance and/or the vehicle is not moving, the robotic/mechanical arm may automatically place the nozzle within the fuel tank opening and provide fuel to the vehicle **12**.

(94) On the other hand, when the fuel tank opening is outside the predetermined maximum distance, the POS terminal **20** may transmit a signal to the vehicle head unit **14** and/or the electronic circuitry **50** requesting that the vehicle **12** be moved to a particular location. For example, when the vehicle **12** is user-operated the POS terminal **20** may transmit a request to move to a particular location which may be presented on the display **236** of the vehicle head unit **14**. In other embodiments, the POS terminal **20** may include speakers and may transmit a voice announcement directing the user to the particular location. In yet other embodiments, the POS terminal **20** may transmit an image of the position of the vehicle **12** with an indication of a position that the vehicle need to move to, such as a rectangular outline. The image may be presented on the display **236** of the vehicle head unit **14**.

(95) When the vehicle **12** is autonomous or semi-autonomous, the request may be transmitted to the vehicle head unit **14** to communicate with the software which controls operation of the vehicle. In any event, the request to move to a particular location may include a request to move the vehicle **12** forward a particular distance, to move the vehicle **12** in reverse a particular distance **12**, to move the vehicle **12** a particular distance toward the fuel pump, and/or a combination of these. When the vehicle **12** moves within the predetermined maximum distance, the robotic arm may automatically place the nozzle within the fuel tank opening and provide fuel to the vehicle **12**. In some embodiments, the autonomous or semi-autonomous vehicle **12** may automatically navigate to the merchant location associated with the POS terminal **20**. The autonomous or semi-autonomous vehicle **12** may also automatically position itself for receiving the goods or services provided by the merchant, such as positioning the autonomous or semi-autonomous vehicle **12** to receive fuel from a nozzle.

(96) In this manner, the entire fueling process may be controlled automatically without requiring the user to leave his car. Instead, the user may pay for and fill his vehicle with fuel simply by selecting controls on the vehicle head unit **14**. Also when the vehicle **12** is autonomous, the autonomous vehicle may travel to the fuel station, receive fuel, and/or pay for the fuel without any assistance from a human.

(97) In addition to making payments from a vehicle, automatically opening and closing a fuel cap and/or communicating with a robotic arm, the electronic circuitry **50** and/or the vehicle head unit **14** may

communicate with a fuel level sensor in the fuel tank. The vehicle **12** may include a fuel level sensor which measures the amount of fuel in the fuel tank. The electronic circuitry **50** and/or the vehicle head unit **14** may be communicatively coupled to the fuel level sensor.

(98) When the user selects the “YES” button **366** to transmit the token for the financial card and/or open the fuel cap of the vehicle **12**, the electronic circuitry **50** and/or the vehicle head unit **14** may measure the amount of fuel in the fuel tank via the fuel level sensor. The electronic circuitry **50** and/or the vehicle head unit **14** may perform an additional measurement of the amount of fuel at the end of the fueling process. Based upon these measurements, the vehicle head unit **14** and/or the electronic circuitry **50** may determine the amount of fuel provided to the vehicle from the fuel pump based upon a change in the amount of fuel before and after the fueling process. As a result, the vehicle head unit **14** and/or the electronic circuitry **50** may estimate a total cost of the fuel based upon the amount of fuel provided to the vehicle. The estimated total cost may be compared to the total cost generated by the POS terminal **20**. When the estimated total cost and the total cost generated by the POS terminal **20** differ by more than a threshold amount (e.g., 10 percent), the vehicle head unit **14** and/or the electronic circuitry **50** may detect a fraudulent charge. Accordingly, the electronic circuitry **50** may transmit an indication that the payment should not be authorized to the vehicle head unit **14**, and/or the vehicle payment application **266** may present a recommendation on the vehicle head unit **14** not to authorize payment for the fuel.

(99) In other embodiments, the electronic circuitry **50** and/or the vehicle head unit **14** may communicate with a fuel level sensor in the fuel tank to identify when the vehicle needs fuel. For example, when the fuel level is below a predetermined threshold (e.g., one-quarter of a tank), the vehicle head unit **14** may determine that the vehicle needs fuel. Accordingly, the vehicle head unit **14** may provide an indication to the user that the vehicle needs fuel and/or may provide a map display including navigation directions to the nearest fuel station which accepts vehicle payments. When the vehicle is an autonomous or semi-autonomous vehicle, the vehicle head unit **14** may automatically provide instructions for the vehicle to navigate to the nearest fuel station and/or a preferred fuel station which accepts vehicle payments. In addition to detecting fraudulent charges and identifying when the vehicle needs fuel, the fuel level sensor may also be used to determine that the fuel tank is full. When the fuel tank is full, the electronic circuitry **50** and/or the vehicle head unit **14** may communicate with the POS terminal **20**, so that the smart fuel pump stops providing fuel.

E. Exemplary Electronic Receipt Screen

(100) After the user authorizes payment for the goods or services and the payment is accepted, an electronic receipt screen **380** as shown in FIG. 3E may be presented by the vehicle payment application **266**. The electronic receipt screen **380** may include information indicative of the payment **382**, including the name of the establishment where the goods or services were purchased (John's Gas Station), the date, the total amount paid, the card type, a masked card number, etc. In some embodiments, the electronic receipt may also be sent to the user's portable device **10** or other computing device, via email, short message service (SMS), etc. Additionally, the POS terminal **20** may cause the goods or services purchased to be provided. In the example above, the POS terminal **20** may instruct the robotic/mechanical arm to place the nozzle within the fuel tank opening and provide fuel to the vehicle **12** by opening a smart valve to allow fuel to flow into the fuel tank and closing the smart valve when the fuel tank is full. In another example, the POS terminal **20** may provide instructions to merchant employees to process a food order, deliver dry cleaning, a prescription, etc. More specifically, the POS terminal **20** may determine that a passenger within the vehicle **12** has a prescription that needs to be filled or delivered to the passenger.

IV. Exemplary Flow Diagram for Vehicle Payment System

(101) FIG. 4 depicts a flow diagram representing an exemplary method **400** for using a vehicle as a payment device. The method **400** may be executed on the vehicle head unit **14**, the electronic circuitry **50** proximately attached to the exterior of the vehicle, and/or a combination of these devices. In some embodiments, a portion of the method **400** may be implemented in a set of instructions stored on a non-transitory computer-readable memory and executable on one or more processors of the vehicle head unit **14**. For example, a portion of the method **400** may be performed by the vehicle payment application **266** of FIG. 2A. Another portion of the method **400** may be implemented in another set of instructions stored on another non-transitory computer-readable memory and executable on one or more processors of the electronic circuitry **50**.

(102) At block **402**, the vehicle payment application **266** may receive financial data for a financial card or financial account, such as a cardholder name, an expiration date, a card number, and/or a CSC code. For example, a user may select a user control such as an “Add Credit/Debit Card” button on a home screen (not

shown) and/or a financial card selection screen of the vehicle payment application **266**. The vehicle payment application **266** may then capture an image of the financial card using a camera in the vehicle head unit **14**, receive manually entered financial data for the financial card from the user, import the financial card from another application such as a photo library, email application, etc., and/or an external source such as a portable device **10** in communication with the vehicle head unit **14**, etc.

(103) The vehicle payment application **266** may obtain a token representing the financial data for the financial card (block **404**). In some embodiments, to obtain a token the vehicle payment application **266** may transmit the financial data to a third-party token server **104** as shown in FIG. **1**, which may in turn generate and/or store the token with the financial data that the token represents. Moreover, the third-party token server **104** may transmit the token to the vehicle payment application **266**.

(104) At block **406**, the vehicle payment application **266** may receive a request to initiate a payment process. For example, the user may select a user control from a payment activation screen of the vehicle payment application **266** indicating the user would like to make a payment from her vehicle. In some embodiments, the payment activation screen may be displayed automatically upon detecting the presence of a wireless signal transmitted by a POS terminal **20**. In other embodiments, the user may select a vehicle payment application icon to display the payment activation screen.

(105) In any event, the vehicle payment application **266** may authenticate the user to ensure she is authorized to use stored financial cards to make payments (block **408**). The vehicle payment application **266** may store biometric information for authorized users, a unique identifier for authorized users, and/or any other suitable information for identifying authorized users. Moreover, the vehicle payment application **266** may request the user to submit a biometric identifier, for example by capturing an image of the user's face and/or by pressing a finger on the display **236** of the vehicle head unit **14**. The vehicle payment application **266** may also request the user to enter login information, such as a username and password, a PIN number, etc. When the biometric identifier and/or login information for the user matches the stored biometric information and/or unique identifier for authorized users, the user may be authenticated. Otherwise, the vehicle payment application **266** may continue to receive biometric identifiers and/or login information from the user to find a match.

(106) When the user is authenticated, the vehicle payment application **266** may receive a selection of a financial card (block **410**), for example via a financial card selection screen. The financial card selection screen may include indications of one or several financial cards stored at the vehicle payment application **266**, and the user may select one of the financial cards displayed on the financial card selection screen.

(107) Upon selecting a financial card and/or authorizing payment using the selected financial card, the vehicle payment application **266** may transmit a token representing the financial data for the financial card to electronic circuitry **50** proximately attached to the exterior of the vehicle **12** (block **412**). The token may be transmitted via a first short-range communication link. For example, the vehicle head unit **14** and the electronic circuitry **50** may be paired using Bluetooth, and the data may be transmitted over the Bluetooth communication protocol. In other embodiments, the data may be transmitted over any other suitable short-range communication link, such as Wi-Fi, USB, DSRC, RFID, etc. Also in some embodiments, the vehicle payment application **266** may have transmitted the token for the financial card to the electronic circuitry **50** when the financial card was initially added to the vehicle payment application **266**. When the user selects the financial card, the vehicle payment application **266** may transmit an indication of the selected financial card to the electronic circuitry **50**.

(108) The electronic circuitry **50** may then transmit the token for the financial card to a POS terminal **20** for making the payment (block **414**). The data may be transmitted over a second very short-range communication link, such as NFC, HF RFID, etc. While the method **400** includes transmitting the token for the financial card from the vehicle head unit **14** to the POS terminal **20** via electronic circuitry **50** proximately attached to the exterior of the vehicle, this is merely an exemplary embodiment. In other embodiments, the token may be transmitted directly from the vehicle head unit **14** to the POS terminal **20**.

(109) In any event, upon receiving the token, the POS terminal **20** may process the payment from the user. If the payment is accepted, the POS terminal **20** may transmit an electronic receipt (block **416**) to be presented on the display **236** of the vehicle head unit **14**. Additionally, the electronic receipt may be transmitted to a portable device **10** of the user via email, SMS, etc. On the other hand, if the payment is declined, the POS terminal **20** may transmit a message to be presented on the display **236** of the vehicle head unit **14** indicating that the selected financial card has been declined and/or to select a different method of payment. The method may include additional, fewer, or alternative actions, including those discussed elsewhere herein.

V. Exemplary Communication Flow for Vehicle Payments

(110) FIG. 5 depicts a connected vehicle **12** configured for use as a payment device, and an exemplary wireless communication flow **500** from a connected vehicle to various merchants having merchant communication terminals configured to accept vehicle pay. Virtual vehicle pay (or vehicle payments) may be used to purchase various goods, such as fast food, dry cleaning, coffee, tolls, car washes, drive thru pharmacy, emergency room services, and/or routine vehicle maintenance.

(111) As shown, a standard purchase flow **510** may include (1) presenting price and/or product information **512**, such as on a vehicle central console, vehicle navigation unit, or infotainment system. The price/product information may be transmitted to the vehicle from a merchant communication terminal or server via wireless communication and/or data transmission over a radio frequency link. In one embodiment, the price and/or product information may be transmitted over a short-range wireless communication channel. After which, a driver or passenger may click or touch an “OK” or other icon on a display screen of a vehicle infotainment system or the like to authorize purchase.

(112) The standard purchase flow **510** may include (2) transmitting financial payment data **514** from the vehicle to the merchant communication terminal or server. The financial payment data may include information related to the name of the driver or passenger, their billing address, CSV, a tokenized card number (such as a credit or debit card), and/or one or more tokens or encryption or other keys used to facilitate secure financial transactions.

(113) The standard purchase flow **510** may include (3) providing an electronic receipt **516** for goods or services purchased via the vehicle payment. The receipt may be emailed to the driver or passenger's mobile device, or stored in a cloud location accessible by the vehicle or a mobile device.

(114) Also shown in FIG. 5 is an exemplary gas purchase flow **520**. The gas purchase flow **520** may include (1) sending or transmitting financial payment data **522** to a merchant or gas station communication terminal or server via wireless communication or data transmission over a radio frequency link. In one embodiment, a short-range wireless communication channel may be used. The financial payment data may include a tokenized card number, name, billing address, CSV, and/or tokens or encryption keys to enable secure financial transactions.

(115) The gas purchase flow **520** may include (2) the merchant or gas station communication terminal or server transmitting price and/or product information **524** before and/or after fueling the vehicle. The price/product information may include price per gallon of gasoline and/or number of gallons to be purchased, or that have already been pumped. The price and/or product information may be transmitted over the short-range wireless communication channel.

(116) The gas purchase flow **520** may include (3) the driver or passenger being presented with the price and/or product information **526**, such as via a display of a vehicle infotainment system, vehicle navigation system, or central control console. After the user touches an “OK” icon or the like, the vehicle may transmit, such as over the short-range wireless communication channel, an authorization to the gas station communication terminal to charge an amount indicated to pay for the gasoline purchased.

(117) The gas purchase flow **520** may include (4) the gas station communication terminal or server transmitting an electronic receipt **528** detailing the purchase of gasoline. The receipt may be, for example, emailed to the user's mobile device, to the vehicle, or sent to a mail box located or accessible via the cloud.

(118) In one embodiment, the driver or passenger financial information (or an account associated with the vehicle itself) (or even driver or passenger biometric data stored in database, with the driver's or passenger's permission or affirmative consent) may be required to be received and verified by the gas station communication terminal, such as a “smart” gas pump, before the smart gas pump will be activated, or otherwise pump or transfer gasoline to the vehicle's gas tank.

(119) In another embodiment, upon entering the vehicle, the driver or passenger may be authenticated by one or more of the following: PIN, voice recognition, facial scan, finger print scan, retina scan, authenticated key fob, presence and/or identification of a mobile device (e.g., smart phone, smart watch, or wearable electronics). The data may be transmitted from the vehicle to a merchant computer or terminal by RFID, DSRC powered signals, Bluetooth low energy, or Wi-Fi signals. The primary driver interface may be the infotainment system of the vehicle, vehicle navigation system, and/or an autonomous vehicle controller or control system, for examples.

(120) Additional features may include the ability of the vehicle communication system/controller to cap spending for each user, each user account, and/or a vehicle account (such as virtual account established for an autonomous vehicle to insure that the autonomous vehicle has access to, or is authorized a certain amount

of monies to pay for gasoline or vehicle maintenance). For instance, a parent may cap a teen's account at a given level. Payment methods may include the use of toggle buttons.

(121) A link to merchant loyalty reward programs may be associated with the various user or vehicle financial accounts used. Other links may be provided, such as links to insurance provider accounts, banking or debit accounts, and/or links to provide medical insurance or health insurance provider and account information upon a visit to a doctor, hospital, clinic, or emergency room.

VI. Exemplary Smart Wireless Communication Terminal

(122) In one aspect, a computer system configured to use a connected vehicle as an electronic payment device may be provided. The computer system may include one or more merchant communication terminal-mounted processors, sensors, biometric sensors and/or transceivers configured to: (1) identify an autonomous or other vehicle (such as via license plate scan); (2) detect a triggering event; (3) generate an electronic message asking the driver or passenger of the vehicle if they would like to authorize payment from a virtual account for nearby goods or services, the virtual account associated with, or linked with, the driver or passenger, and/or the autonomous or other vehicle, and transmitting the electronic message via wireless communication or data transmission over a radio link to the vehicle; (4) receive via wireless communication or data transmission over the radio link a response an input from the driver or passenger authorizing payment for the goods or services, such as from a vehicle account or a virtual account associated with the driver or passenger; (5) securely transfer funds from the virtual account to a merchant's virtual account or merchant communication terminal (such as via wireless communication or data transmission over a short-range wireless communication channel); and/or (6) transmit an order confirmation (such as via wireless communication or data transmission over a short-range wireless communication channel), and/or move the goods or services into or in proximity of the vehicle to facilitate the vehicle acting as a payment device.

(123) In another aspect, a smart wireless communication terminal for facilitating using a connected vehicle as a payment device may be provided. The smart wireless communication terminal may be associated with a merchant that provides goods or services, and may include one or more processors and/or transceivers configured to: (1) detect that a connected vehicle is within direct wireless communication or data transmission range, the connected vehicle being equipped with one or more processors or transceivers and configured for short-range wireless communication over a radio link; (2) establish a secure or encrypted wireless connection over a secure short-range communication network with the connected vehicle; (3) electronically verify an identification of an individual riding within the connected vehicle; (4) electronically verify a virtual financial account associated with the individual riding within the connected vehicle; (5) accept a virtual order for goods or services from the connected vehicle input from the individual; and/or (6) automatically deliver the goods or services orders (e.g., food or gas) into or to the vicinity of the connected vehicle, or generate a notification that payment for the goods or services has been accepted/received, to facilitate using the vehicle as a payment device and to complete commercial transactions.

(124) In yet another aspect, a smart wireless communication terminal for facilitating using a vehicle as a payment device may be provided. The smart wireless communication terminal may be associated with a merchant that provides goods or services. The smart wireless communication terminal may include one or more processors and/or transceivers configured to: (1) detect that a connected vehicle is within direct wireless communication or data transmission range, the connected vehicle being equipped with one or more processors or transceivers and configured for short-range wireless communication over a radio link; (2) establish a secure or encrypted wireless connection over a secure short-range communication network with the connected vehicle; (3) electronically verify an identification of the connected vehicle, such as via the secure or encrypted wireless communication or via a license plate scan or image analysis of digital images of the license plate; (4) electronically verify a virtual financial account associated with an individual riding within the connected vehicle, or with the connected vehicle itself; (5) accept a virtual order for goods or services from the connected vehicle input from the individual; and/or (6) automatically deliver the goods or services orders (e.g., food or gas) into or to the vicinity of the connected vehicle, or generate a notification that payment for the goods or services has been accepted/received, to facilitate using the vehicle as a payment device and to complete commercial transactions.

(125) The foregoing smart merchant or wireless communication terminals may be further configured to automatically pump or push gas into an open gas tank of the connected vehicle; to automatically detect that the connected vehicle is in position to receive gas, or to receive an automated, mechanical, or robotic arm associated with a smart nozzle or pump that facilitates filling the gas tank of the connected vehicle; to securely accept payment from the virtual account for the goods or services; and/or to sense a location of the

connected vehicle with respect to the smart terminal. The terminal may be further configured to sense a location of the gas tank opening of the connected vehicle with respect to the smart terminal, and/or whether the gas tank opening is open (such as whether a smart gas tank cover has automatically opened to allow a robotic arm to move a nozzle into the opening and allow passage of gasoline into the connected vehicle's gas tank). Additionally or alternatively, the smart wireless communication terminal may be further configured to automatically extend the robotic arm and/or nozzle into a position that allows the robotic arm, nozzle, and/or smart terminal to fill the gas tank of the connected vehicle.

(126) In another aspect, a smart communication terminal for facilitating using a vehicle as a payment device may be provided. The smart communication terminal may be associated with a merchant that provides goods or services, and may include (1) means for detecting that a connected vehicle is within direct wireless communication or data transmission range, the connected vehicle being equipped with one or more processors or transceivers and configured for wireless communication over one or more radio links; (2) means for establishing a secure or encrypted wireless connection over a secure communication network with the connected vehicle; (3) means for electronically verifying an identification of an individual riding within the connected vehicle; (4) means for electronically verifying a virtual account associated with the individual riding within the connected vehicle; (5) means for accepting a virtual order for goods or services from the connected vehicle input from the individual; and/or (7) means for automatically delivering the goods or services orders (e.g., food or gas) into, or to the vicinity of, the connected vehicle to facilitate using the vehicle as a payment device and to complete commercial transactions.

(127) The smart communication terminal may further include (8) means for automatically pumping or pushing gas into an open gas tank of the connected vehicle; (9) means for automatically detecting that the connected vehicle is in position to receive gas, or to receive an automated, mechanical, or robotic arm associated with a smart nozzle that facilitates filling the gas tank of the connected vehicle; (10) means for securely accepting payment from the virtual account for the goods or services; (11) means for sensing location of the connected vehicle with respect to the smart terminal; (12) means for sensing location of the gas tank opening of the connected vehicle with respect to the smart terminal, and/or whether the gas tank opening is open (such as whether a smart gas tank cover has automatically opened to allow a robotic arm to move a nozzle into the opening and allow passage of gasoline into the connected vehicle's gas tank); and/or (13) means for automatically extending the robotic arm and/or nozzle into a position that allows the robotic arm, nozzle, and/or smart terminal to fill the gas tank of the connected vehicle. The smart communication terminal may include additional, less, or alternate functionality, including that discussed elsewhere herein.

(128) The (1) means for detecting that a connected vehicle is within direct wireless communication or data transmission range may include the connected vehicle being equipped with one or more processors or transceivers and configured for wireless communication over one or more radio links. The one or more processors and/or transceivers may be able to detect a low energy transmitter (such as a Bluetooth transmitter or Wi-Fi transmitter) located on the connected vehicle, or a low energy short-range communication signal or channel. Additionally or alternatively, the one or more processors may compare a vehicle's current GPS location that is received via wireless communication with a stored GPS location of the merchant communication terminal (such as a smart gas pump) to determine that short-range and/or low energy wireless communication from the merchant communication terminal to the connected vehicle is possible because they are within a few feet of each other (assuming both are in working condition).

(129) The (2) means for establishing a secure or encrypted wireless connection over a secure communication network with the connected vehicle may include the one or more processors and/or transceivers of the merchant communication terminal and/or vehicle transmitting one or more encryption keys or tokens to each other for unlocking a subsequent transmission, or otherwise transmitting an encrypted wireless communication or data transmission from the merchant communication terminal.

(130) The (3) means for electronically verifying an identification of an individual riding within the connected vehicle may include the merchant communication terminal receiving an identification of the passenger or driver (and/or vehicle) from the vehicle transmitter after the vehicle controller verifies identification of the passenger or driver, such as via one or more biometric sensors. Additionally or alternatively, the biometric sensor data may be transmitted to the merchant communication terminal for verification of driver or passenger identity, such as via comparison with stored driver or passenger biometric data (stored with the driver's or passenger's permission or affirmative consent).

(131) The (4) means for electronically verifying a virtual account associated with the individual riding within

the connected vehicle may include one or more processors and/or transceivers of the merchant communication terminal verifying with a 3rd party server, such as a remote server associated with a financial institution, that the virtual account exists and is associated with the individual and/or vehicle identified.

(132) The (5) means for accepting a virtual order for goods or services from the connected vehicle input from the individual may include the one or more processors and/or transceivers associated with the merchant communication terminal receiving the virtual order from the connected vehicle controller/transceiver via the short-range, low energy wireless communication channel.

(133) The (7) means for automatically delivering the goods or services orders (e.g., food or gas) into or to the vicinity of the connected vehicle may include automatically directing the vehicle, such as an autonomous vehicle, to position itself to a drive-up window (for food, snacks, coffee, prescription drugs, dry cleaning, etc.), or to a smart gas pump. The vehicle may be directed using GPS location data or coordinates of the vehicle, the drive-up window or smart gas pump, and a known or determined layout of a merchant parking lot.

(134) The (8) means for automatically pumping or pushing gas into an open gas tank of the connected vehicle may include a smart gas pump configured with a robotic or mechanic arm that automatically extends from the smart gas pump toward the vehicle to move a nozzle into an open gas tank of the vehicle, and to allow the initiating of pumping gasoline into the vehicle. The nozzle may be a dumb or smart nozzle, and/or the gas tank may include a smart gas tank cover that opens and/or shuts under the direct of commands received via wired or wireless communication from the connected vehicle controller or via wireless communication from a merchant communication terminal or a smart gas pump configured for wireless communication.

(135) The (9) means for automatically detecting that the connected vehicle is in position to receive gas, or to receive an automated, mechanical, or robotic arm associated with a smart (or other) nozzle that facilitates filling the gas tank of the connected vehicle may include the smart communication terminal or smart gas pump including one or more sensors, cameras, infrared devices, radar units, or lights that measure the distance to the vehicle and/or the height of the gas tank of the vehicle (or determines a make/model of the vehicle, and then looks height of the gas tank from a data structure). The one or more sensors, cameras, infrared devices, radar units or lights may be configured to detect the gas tank cover is open or that a smart gas tank cover has opened. Alternatively, the vehicle may be equipped with a smart gas tank cover that wirelessly communicates with the vehicle controller and/or the merchant communication terminal indicating that the smart gas tank cover has opened or is open (such as by sliding, moving, or rotating a cover or flap).

(136) The (10) means for securely accepting payment from the virtual account for the goods or services may include the one or more processors being able to use (or receive) one or more tokens or encryption keys to securing access the financial account of the users or the vehicle.

(137) The (11) means for sensing location of the connected vehicle with respect to the smart merchant communication terminal may include one or more sensors, radar units, cameras, light sensors unit, 3D image units mounted on the smart merchant terminal, and/or the one or more processors and/or transceivers mounted on the smart merchant terminal being in wireless communication with the vehicle, and/or configured to request and receive the vehicle's current GPS location.

(138) The (12) means for sensing location of the gas tank opening of the connected vehicle with respect to the smart terminal, and/or whether the gas tank opening is open (such as whether a smart gas tank cover has automatically opened to allowed a robotic arm to move a nozzle into the opening and allow passage of gasoline into the connected vehicle's gas tank) may include one or more sensors, cameras, radar units, light reflection measuring units, 3D imaging units, or other sensors configured to measure the distance from the smart merchant terminal (or smart gas pump) to the vehicle, and/or to determine or identify the height of the gas tank opening, and/or whether the gas tank is open (such as a smart gas tank cover being opened automatically via control signal sent from the vehicle (or vehicle controller) or a wireless control signal sent from the smart merchant communication terminal (or gas pump)).

(139) The (13) means for automatically extending the robotic arm and/or nozzle into a position that allows the robotic arm, nozzle, and/or smart terminal to fill the gas tank of the connected vehicle may include a robotic or mechanical arm that physical expands or extends horizontally from the smart terminal toward the gas tank of the connected vehicle until the nozzle is positioned within the open gas tank opening to allow the smart terminal to commence pumping gas and to fill up the gas tank. The pumping of gas may be commenced once the smart terminal or vehicle verify that the nozzle is positioned within the opening of the gas tank, such as via one or more contact sensors, a sensor array, or via digital image analysis. The smart

terminals mentioned herein may include additional, less, or alternate functionality, including that discussed elsewhere herein.

(140) In another aspect, a computer-implemented method for using a point-of-sale terminal to accept vehicle payments may be provided. The method may include: (1) receiving (via one or more processors (and/or associated transceivers) in a POS terminal) an indication that a vehicle for providing electronic payments is within a predetermined range of the POS terminal; (2) transmitting (via the one or more processors and/or associated transceivers, and/or a wired or wireless communication and/or data transmission over a radio link or wireless communication channel) an electronic message including an indication of goods or services that can be purchased at the POS terminal and a user control for transmitting a request to order the goods or services to the POS terminal, wherein the electronic message is displayed on a head unit of the vehicle; in response to receiving an indication from the vehicle that the vehicle or a user within the vehicle wants to initiate a payment process; and/or (3) receiving (via the one or more processors and/or associated transceivers, and/or a wired or wireless communication and/or data transmission from an electronic device proximately attached to an exterior of the vehicle over a radio link or wireless communication channel) information indicative of a financial account for making payments. The method may further include: (4) providing (via the one or more processors) the goods or services to the vehicle; (5) transmitting (via the one or more processors and/or associated transceivers, and/or a wired or wireless communication and/or data transmission over a radio link or wireless communication channel) the information indicative of the financial account to an issuing bank server for processing payment for the goods or services; and/or when the payment is accepted as indicated by the issuing bank server, (6) transmitting (via the one or more processors and/or associated transceivers, and/or a wired or wireless communication and/or data transmission over a radio link or wireless communication channel to the vehicle) an electronic receipt indicative of the payment to facilitate using the vehicle as a payment device.

(141) The foregoing methods may include additional, less, or alternate actions, including those discussed elsewhere herein. The foregoing methods may be implemented via one or more local or remote processors, sensors, and/or transceivers, and/or via computer-executable instructions stored on non-transitory computer-readable medium or media.

(142) For instance, providing the goods or services to the vehicle may include providing at least one of: gasoline, food, dry cleaning, coffee, groceries, pharmaceuticals, or vehicle maintenance to the vehicle. The goods or services may be gasoline and the method may further include: (1) detecting (via the one or more processors) that a gas tank of the vehicle is open and a position of a gas tank opening for the vehicle; (2) directing (via the one or more processors) a robotic arm included in a smart gas pump communicatively coupled to the POS terminal to insert a nozzle attached to the robotic arm into the gas tank opening according to the position of the gas tank opening; (3) automatically providing (via the one or more processors) gasoline to the vehicle by causing a smart valve within the smart gas pump to open; (4) detecting (via the one or more processors) that the gas tank of the vehicle is full; and/or (6) closing (via the one or more processors) the smart valve to stop providing gasoline to the vehicle. The method may further include: detecting (via the one or more processors) that the gas tank opening is further than a predetermined maximum distance away from the POS terminal according to the position of the gas tank opening; and/or transmitting (via the one or more processors and/or associated transceivers, and/or a wired or wireless communication and/or data transmission over a radio link or wireless communication channel to the vehicle) a request that the vehicle be moved to a particular location which is less than the predetermined maximum distance from the POS terminal.

(143) Additionally, the POS terminal may direct the robotic arm to insert the nozzle into the gas tank opening when the position of the gas tank opening is less than a predetermined maximum distance away from the POS terminal. The electronic message may be transmitted via the very short-range communication link to the electronic device proximately attached to the exterior of the vehicle, and the electronic device may transmit the electronic message, via a short-range communication link, to the head unit of the vehicle for display.

(144) Moreover, the method may also include establishing (via the one or more processors) the very short-range communication link between the POS terminal and the electronic device proximately attached to the exterior of the vehicle for receiving electronic payments when the vehicle is within the predetermined range of the POS terminal.

(145) In yet another aspect, a merchant communication terminal for facilitating using a vehicle as a payment device may be provided. The merchant communication terminal may include one or more processors (and/or associated transceivers) in a merchant communication terminal associated with a merchant, and/or a non-

transitory computer-readable memory coupled to the one or more processors and storing machine readable instructions, that when executed by the one or more processors, may cause the merchant communication terminal to perform various tasks. For example, the instructions may cause the system to: (1) detect a triggering event indicating that a vehicle is within a predetermined range for communicating with the merchant communication terminal; (2) establish a short-range communication link between the merchant communication terminal and the vehicle; (3) electronically verify identification information indicative of the vehicle or a user within the vehicle; (4) electronically verify a virtual account associated with the vehicle or the user within the vehicle; (5) accept a virtual order for goods or services provided by the merchant; and/or (6) automatically provide the goods or services to the vehicle to facilitate using the vehicle as a payment device. The merchant communication terminal may include additional, fewer, or alternate components and/or functionality, including that discussed elsewhere herein.

(146) For instance, the goods or services provided by the merchant may be gasoline and the merchant communication terminal may be communicatively coupled to a smart gas pump having a robotic arm for maneuvering a nozzle to automatically pump gas into a gas tank opening of the vehicle. Identification information indicative of the vehicle or a user within the vehicle may include a PIN number, a facial scan, a finger print scan, a retina scan, a voice scan, an authenticated key fob, mobile device identification information, sensor data from weight or pressure sensors within the vehicle which detect a weight of a driver or passenger within the vehicle, or a license plate number for the vehicle.

(147) Additionally, to detect a triggering event indicating that a vehicle is within a predetermined range for communicating with the merchant communication terminal, the instructions may cause the merchant communication terminal to at least one of: detect that the vehicle is within Wi-Fi, Radio Frequency Identification (RFID), or Bluetooth communication range of the merchant communication terminal by receiving a Wi-Fi, RFID, or Bluetooth signal from the vehicle; and/or detect that the vehicle is within a predetermined distance of the merchant communication terminal by capturing images, via a camera communicatively coupled to the merchant communication terminal, of an area surrounding the merchant communication terminal, and/or identifying, using digital image analysis techniques, the vehicle within at least one of the images and a position of the vehicle relative to the merchant communication terminal.

(148) Moreover, the instructions may further cause the merchant communication terminal to: (1) generate an electronic message requesting a user to authorize payment for the goods or services from the virtual account associated with the vehicle or the user; (2) transmit, via the short-range communication link, the electronic message to the vehicle; (3) receive, via the short-range communication link, input from the user authorizing payment from the goods or services from the virtual account; (4) detect that a gas tank of the vehicle is open and a location of the gas tank opening for the vehicle with respect to the merchant communication terminal; (5) direct the robotic arm included in the smart gas pump to insert the nozzle into the gas tank opening according to the position of the gas tank opening; (6) automatically provide gasoline to the vehicle; (7) accept payment from the virtual account for the goods or services; and/or (8) transmit, via the short-range communication link, an electronic receipt for the payment of the goods or services.

VII. Exemplary Smart Gas Pump

(149) FIG. 6 depicts an exemplary system that is configured to facilitate using a connected vehicle as a payment device **600**. A smart gas pump **602** may include a communication terminal **614** that includes one or more processors and/or transceivers, similar to the POS terminal **20**, as shown in FIG. 1 and described above. The communication terminal **614** may be configured for short-range, very short-range, and/or low energy wireless communication or data transmission (such as Wi-Fi, Bluetooth, RFID, NFC). The communication terminal **614** may be in wireless communication with a vehicle infotainment, vehicle controller, or vehicle navigation system **616** mounted within the vehicle **610**, which may be a smart vehicle or an autonomous vehicle in some embodiments. The communication terminal **614** associated with the merchant may communicate with the vehicle **610** over a short-range, very short-range, and/or low energy radio frequency link or channel **618**.

(150) The smart gas pump **602** may include one or more sensors (such as cameras, 3D imaging devices, radar units, light emitting devices, etc.), such as within a sensor array **609**. The sensor array **609** may be configured to detect the location of the vehicle, type of vehicle (such as by employing object recognition, optical character recognition, or pattern recognition techniques on digital images of the vehicle to determine make, model, year, and/or license plate number), and/or a location and/or height of the gas tank of the vehicle. The sensor array **609** may also be able to detect whether a gas tank opening **612** of the vehicle **610** is open or otherwise not covered. Additionally or alternatively, the vehicle **610** may have sensors that indicate

that the gas tank is open or otherwise ready to accept a nozzle, and may communicate that fact with the smart gas pump **602** or communication terminal **614**.

(151) Additionally or alternatively, the vehicle **610** may be equipped with a smart gas tank cap or cover that is equipped with a transceiver that communicates its status (open or closed) to the smart gas pump **602** or communication terminal **614**. In some embodiments, the smart gas pump **602** or communication terminal **614** may send a control signal for a smart gas tank cover to open once the sensor array **609** determines that the vehicle **610** is positioned properly to receive gas from the smart gas pump.

(152) The smart gas pump **602** may include a robotic or mechanical arm **604** that is equipped with a smart or other nozzle **608**, and/or smart valve or handle **606**. The smart nozzle **608** may include a sensor that detects and/or indicates whether the nozzle **608** is positioned correctly within a gas tank opening to pump or otherwise allow gasoline to move into a vehicle gas tank. The smart valve or handle **606** may open and/or allow gasoline to move into the vehicle gas tank after being directed to open by the communication terminal **614** and/or the smart nozzle **608**.

(153) The communication terminal **614** may be in communication with, and/or send control signals to the smart nozzle **608**, smart valve **606**, and/or robotic or mechanical arm **604**. For instance, once the sensor array **609** determines that the vehicle **610** is parked and in a position to receive gasoline (and the gas tank is open or uncovered), that information may be received by the communication terminal **614**. After which, the communication terminal **614** may direct the robotic or mechanical arm **604** to extend from the smart gas pump **602** at a distance and/or height determined from the sensor array **609** that is appropriate for the vehicle's make/model. Once the arm **604** is extended and the smart nozzle **608** sensor or other sensor indicates that the smart nozzle **608** is positioned within the gas tank opening, the communication terminal **614** may direct the smart valve **606** to open to allow gasoline to flow into the gas tank—until another sensor determines that the gas tank is approximately or almost full.

(154) For instance, FIG. 7 shows the robotic or mechanical arm **704** in an extended position. The smart gas pump **702** may include one or more processors or a communication terminal **716** configured for wired and/or wireless communication. The smart gas pump **702** may include the fore-mentioned or another sensor array **718**. The arm **704** may include a smart or dumb nozzle **708**, and a means for initiating/stopping the flow of gasoline into the vehicle **710**, such as a smart valve or handle **706**. The arm **704** may be extended after the sensor array **718** determines that the gas tank opening **712** of the vehicle **710** is uncovered and the vehicle is within distance. For instance, a smart gas tank cover **714** may be directed to swing open by the vehicle **710** controller, or by the smart gas pump **702** communication terminal **716**.

VIII. Exemplary Merchant Smart Communication Terminal Flow

(155) FIG. 8 depicts an exemplary computer-implemented method of conducting commercial transactions with an autonomous vehicle configured for vehicle pay via merchant terminals **800**. The method **800** may include (1) detecting or determining a license plate number of an autonomous vehicle **802**. For instance, when an autonomous vehicle pulls up to a gas station pump or drive-thru window of pharmacy or restaurant, the merchant may be equipped with one or more cameras that acquire digital images of an autonomous vehicle. Object recognition or optical character recognition techniques may be performed on the digital images to determine the license plate number of the autonomous vehicle.

(156) The method **800** may include (2) verifying that the autonomous vehicle is not reported stolen **804**. For instance, from the digital images, the license plate and make/model information may be determined by a merchant terminal. The merchant terminal may access a 3.sup.rd party database associated with law enforcement to verify that the autonomous vehicle has not been reported stolen.

(157) The method **800** may include (3) detecting a trigger event at the merchant terminal **806**. Example events may include the merchant terminal detecting (i) that the autonomous vehicle is in short-range or low-energy wireless communication range, and/or (ii) that the autonomous vehicle has sent an electronic request for goods/services via wireless communication or data transmission. In response, the merchant terminal may transmit product and price information to the autonomous vehicle (such as to the infotainment system), such as an online menu for food/drinks.

(158) The method **800** may include (4) receiving vehicle and/or passenger financial account information from the autonomous vehicle at the merchant terminal **808**. For instance, the vehicle or passenger may have an account with a merchant that they prefer or use regularly, such as an account with a preferred or normal pharmacy, gas station, or restaurant chain. Additionally or alternatively, the merchant terminal may determine vehicle or passenger account information from the license plate determined via image analysis. The merchant terminal may also verify that a prescription has been prescribed by a physician or hospital for

the passenger.

(159) The method **800** may include, under the direction and/or control of the merchant terminal (and/or one or more autonomous vehicle processors), (5) determining a location of a gas tank opening of the autonomous vehicle via a sensor array associated with the merchant terminal **810**; (6) automatically receiving or delivering the goods or services, and/or direct or move the autonomous vehicle to receive the goods or services ordered, such as automatically move the autonomous vehicle through a drive-thru or up to a gas pump **812**; and/or (7) receive e-payment and/or transmit an e-receipt to the autonomous vehicle **814**, such as discussed elsewhere herein.

(160) In one aspect, a computer-implemented method of using a connected vehicle as an electronic payment device may be provided. The method may include (1) identifying or determining, via one or more (merchant smart communication terminal) processors and/or sensors, an identification of a connected autonomous vehicle (such as via object recognition performed on license plate digital images of the vehicle) and/or make and model information; (2) verifying, via the one or more processors and/or transceivers, that the connected autonomous vehicle has not been reported stolen (such as by sending the license plate number and make/model information to a 3.sup.rd party server via wireless communication and/or data transmission over one or more radio links); (3) detecting, via the one or more processors and/or transceivers, a triggering event (such as the connected autonomous vehicle being within short-range communication range with a merchant communication terminal, and/or receiving an electronic request for goods and services from the autonomous vehicle via wireless communication or data transmission); (4) receiving, via the one or more processors (and/or transceivers), vehicle and/or passenger financial virtual account information (such as via wireless communication or data transmission from the autonomous vehicle using a short-range, low-energy communication channel) or otherwise determining a virtual account from the license plate number; (5) accepting, via the one or more processors, an input from the driver or passenger authorizing payment for the goods or services (such as via wireless communication or data transmission from the autonomous vehicle using a short-range, low-energy communication channel); (6) securely, via the one or more processors and/or transceivers, transferring funds from the virtual account to a merchant's virtual account (such as via wireless communication with the merchant communication terminal over a short-range communication channel); and/or (7) moving, via the one or more processors, the goods or services into proximity of the autonomous vehicle; (8) directing, via the one or more processors, the autonomous vehicle into position to receive the goods or services (such as via wireless communication or data transmission sent to the autonomous vehicle using a short-range, low-energy communication channel); (9) confirming, via the one or more processors, delivery of the goods or services (such as via digital image analysis or machine learning techniques performed on images taken of the autonomous vehicle); and/or (10) transmitting a notification that electronic payment for the goods or services (such as via wireless communication or data transmission sent to the autonomous vehicle using the short-range, low-energy communication channel) has been accepted to facilitate the autonomous vehicle acting as an payment device for a commercial transaction.

(161) The method may include detecting, via the one or more processors and/or sensors, a height and distance to the autonomous vehicle gas tank opening (for a gas station merchant terminal); determining, via the one or more processors, and/or sensors, that the gas tank is open or otherwise uncovered; and/or directing, via the one or more processors, sensors, and/or transceivers, the automatic filling of the autonomous vehicle gas tank with gasoline once it is verified that the gas tank is uncovered, and the height and/or distance to the gas tank opening.

(162) The method may include detecting, verifying, or determining, via the one or more processors and/or transceivers, that a passenger within the autonomous vehicle has a prescription that needs to be filled and/or delivered to the passenger (for a pharmacy merchant terminal). The foregoing methods may include additional, less, or alternate actions, including those discussed elsewhere herein.

(163) In another aspect, a computer system or merchant terminal configured to use a connected vehicle as an electronic payment device may be provided. The computer system or merchant terminal may include one or more processors, sensors, cameras, and/or transceivers configured to: (1) identify or determine an identification of a connected autonomous vehicle (such as via object recognition performed on license plate digital images of the vehicle) and/or make and model information; (2) verify that the connected autonomous vehicle has not been reported stolen (such as by sending the license plate number and make/model information to a 3.sup.rd party server via wireless communication and/or data transmission over one or more radio links); (3) detect a triggering event (such as the connected autonomous vehicle being within short-range and/or low energy communication range with a merchant communication terminal, and/or receiving an

electronic request for goods and services from the autonomous vehicle via wireless communication or data transmission); (4) receive vehicle and/or passenger financial virtual account information (such as via wireless communication or data transmission from the autonomous vehicle using a short-range, low-energy communication channel) or otherwise determining a virtual account from the license plate number; (5) accept an input from the driver or passenger authorizing payment for the goods or services (such as via wireless communication or data transmission from the autonomous vehicle using a short-range and/or low-energy communication channel); (6) securely transfer funds from the virtual account to a merchant's virtual account (such as via wireless communication with the merchant communication terminal over a short-range communication channel); (7) move the goods or services into proximity of the autonomous vehicle; (8) direct the autonomous vehicle into position to receive the goods or services (such as via wireless communication or data transmission sent to the autonomous vehicle using the short-range, low-energy communication channel); (9) confirm delivery of the goods or services (such as via digital image analysis or machine learning techniques performed on images taken of the autonomous vehicle); and/or (10) transmit a notification that electronic payment for the goods or services (such as via wireless communication or data transmission sent to the autonomous vehicle using the short-range, low-energy communication channel) has been accepted to facilitate the autonomous vehicle acting as an payment device for a commercial transaction. (164) The one or more processors, sensors, cameras, and/or transceivers may be configured to: detect a height and distance to the autonomous vehicle gas tank opening (for a gas station merchant terminal); determine that the gas tank is open or otherwise uncovered; and/or direct the automatic filling of the autonomous vehicle gas tank with gasoline once it is verified that the gas tank is uncovered, and/or the height and/or distance to the gas tank opening. The one or more processors, sensors, cameras, and/or transceivers may be configured to: detect, verify, or determine that a passenger within the autonomous vehicle has a prescription that needs to be filled and/or delivered to the passenger (for a pharmacy merchant terminal).

(165) The computer-implemented methods discussed herein may include additional, less, or alternate actions, including those discussed elsewhere herein. The methods may be implemented via one or more local or remote processors, transceivers, and/or sensors (such as processors, transceivers, and/or sensors mounted on vehicles or mobile devices, or associated with smart infrastructure or remote servers), and/or via computer-executable instructions stored on non-transitory computer-readable media or medium.

(166) Additionally, the computer systems discussed herein may include additional, less, or alternate functionality, including that discussed elsewhere herein. The computer systems discussed herein may include or be implemented via computer-executable instructions stored on non-transitory computer-readable media or medium.

(167) In yet another aspect, a merchant communication terminal for using an autonomous vehicle as an electronic payment device may be provided. The merchant communication terminal may include one or more processors (and/or associated transceivers) in a merchant communication terminal associated with a merchant, and/or a non-transitory computer-readable memory coupled to the one or more processors and storing machine readable instructions, that when executed by the one or more processors, may cause the merchant communication terminal to perform various tasks. For example, the instructions may cause the system to: (1) obtain identification information for an autonomous vehicle; (2) verify that the autonomous vehicle has not been reported stolen by communicating the identification information to a server for retrieving stolen vehicle records; (3) detect a triggering event indicating that the autonomous vehicle is within a predetermined range for communicating with the merchant communication terminal; and/or (4) receive, via a short-range communication link, virtual account information for a virtual account associated with the vehicle. The instructions may further cause the system to: (5) receive and accept, via the short-range communication link, an input from the vehicle authorizing payment for goods and services provided by the merchant; (6) securely receive, via the short-range communication link, funds from the virtual account at a virtual account for the merchant; and/or (7) transmit, via the short-range communication link, a notification that electronic payment for the goods or services has been accepted to facilitate the autonomous vehicle acting as a payment device. The merchant communication terminal may include additional, fewer, or alternate components and/or functionality, including that discussed elsewhere herein.

(168) For instance, the goods or services may be gasoline, the merchant communication terminal may be communicatively coupled to a smart gas pump having a robotic arm for maneuvering a nozzle to automatically pump gas into a gas tank opening of the vehicle, and the instructions may cause the merchant communication terminal to: (1) determine that a gas tank of the autonomous vehicle is open or uncovered;

(2) determine a height of the gas tank opening of the autonomous vehicle and a distance from the merchant communication terminal to the gas tank opening; and/or (3) direct the robotic arm included in the smart gas pump to insert the nozzle into the gas tank opening for automatic filling of the gas tank. Furthermore, the goods or services may be prescriptions and the instructions may cause the merchant communication terminal to: determine that a passenger within the autonomous vehicle has a prescription that needs to be filled or delivered to the passenger.

(169) Additionally, the instructions may further cause the merchant communication terminal to transfer the goods or services to the autonomous vehicle by: (1) directing the autonomous vehicle to a particular location for receiving the goods or services; and/or (2) confirming delivery of the goods or services by obtaining an indication that the autonomous vehicle received the goods or services.

(170) Moreover, to obtain an indication that the autonomous vehicle received the goods or services, the instructions may cause the merchant communication terminal to at least one of: (1) receive, via the short-range communication link, an electronic message from the autonomous vehicle indicating that the goods or services have been received; and/or (2) capture, via a camera communicatively coupled to the merchant terminal, images of the autonomous vehicle and analyze the images to determine that the goods or services were received.

IX. Additional Considerations

(171) Although the following text sets forth a detailed description of numerous different embodiments, it should be understood that the legal scope of the description is defined by the words of the claims set forth at the end of this patent and equivalents. The detailed description is to be construed as exemplary only and does not describe every possible embodiment since describing every possible embodiment would be impractical. Numerous alternative embodiments may be implemented, using either current technology or technology developed after the filing date of this patent, which would still fall within the scope of the claims.

(172) It should also be understood that, unless a term is expressly defined in this patent using the sentence “As used herein, the term ‘_____’ is hereby defined to mean . . .” or a similar sentence, there is no intent to limit the meaning of that term, either expressly or by implication, beyond its plain or ordinary meaning, and such term should not be interpreted to be limited in scope based upon any statement made in any section of this patent (other than the language of the claims). To the extent that any term recited in the claims at the end of this patent is referred to in this patent in a manner consistent with a single meaning, that is done for sake of clarity only so as to not confuse the reader, and it is not intended that such claim term be limited, by implication or otherwise, to that single meaning. The patent claims at the end of this patent application are not intended to be construed under 35 U.S.C. § 112(f) unless traditional means-plus-function language is expressly recited, such as “means for” or “step for” language being explicitly recited in the claim(s).

(173) Accordingly, the term “vehicle” may refer to any of a number of motorized transportation devices. A vehicle may be a car, truck, bus, train, boat, plane, motorcycle, snowmobile, other personal transport devices, etc.

(174) As used herein, the terms “financial card,” “financial account,” and “virtual account” may be used to refer to any device or identifier that enables the cardholder to make a payment via an electronic transfer of funds. For example, a financial card, financial account, or virtual account may include a credit card, a debit card, a gift card, a charge card, a stored-value card, a rewards card, a bank account, a rewards account, a pre-paid toll account, etc.

(175) The term “point-of-sale (POS) terminal” as used herein may refer to an electronic device used to process financial card payments at retail locations. For example, a POS terminal may include a computer, a cash register and/or other equipment or software for reading financial data, recording transactions, and/or communicating with a credit card network to transfer funds.

(176) As used herein, the term “establishment” may be used to refer to a merchant and/or retail location for selling goods or services to customers. For example, an establishment may include a restaurant, a gas station, a car wash, a clothing store, a department store, a furniture store, a grocery store, a convenience store, a dry cleaning store, a pharmacy, a vehicle repair shop, etc.

(177) The following additional considerations apply to the foregoing discussion. Throughout this specification, plural instances may implement components, operations, or structures described as a single instance. Although individual operations of one or more methods are illustrated and described as separate operations, one or more of the individual operations may be performed concurrently, and nothing requires that the operations be performed in the order illustrated. Structures and functionality presented as separate components in example configurations may be implemented as a combined structure or component.

Similarly, structures and functionality presented as a single component may be implemented as separate components. These and other variations, modifications, additions, and improvements fall within the scope of the subject matter herein.

(178) Additionally, certain embodiments are described herein as including logic or a number of routines, subroutines, applications, or instructions. These may constitute either software (e.g., code embodied on a machine-readable medium or in a transmission signal) or hardware. In hardware, the routines, etc., are tangible units capable of performing certain operations and may be configured or arranged in a certain manner. In example embodiments, one or more computer systems (e.g., a standalone, client or server computer system) or one or more hardware modules of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware module that operates to perform certain operations as described herein.

(179) In various embodiments, a hardware module may be implemented mechanically or electronically. For example, a hardware module may comprise dedicated circuitry or logic that is permanently configured (e.g., as a special-purpose processor, such as a field programmable gate array (FPGA) or an application-specific integrated circuit (ASIC)) to perform certain operations. A hardware module may also comprise programmable logic or circuitry (e.g., as encompassed within a general-purpose processor or other programmable processor) that is temporarily configured by software to perform certain operations. It will be appreciated that the decision to implement a hardware module mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) may be driven by cost and time considerations.

(180) Accordingly, the term “hardware module” should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein.

Considering embodiments in which hardware modules are temporarily configured (e.g., programmed), each of the hardware modules need not be configured or instantiated at any one instance in time. For example, where the hardware modules comprise a general-purpose processor configured using software, the general-purpose processor may be configured as respective different hardware modules at different times. Software may accordingly configure a processor, for example, to constitute a particular hardware module at one instance of time and to constitute a different hardware module at a different instance of time.

(181) Hardware modules may provide information to, and receive information from, other hardware modules. Accordingly, the described hardware modules may be regarded as being communicatively coupled. Where multiple of such hardware modules exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) that connect the hardware modules. In embodiments in which multiple hardware modules are configured or instantiated at different times, communications between such hardware modules may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware modules have access. For example, one hardware module may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware module may then, at a later time, access the memory device to retrieve and process the stored output. Hardware modules may also initiate communications with input or output devices, and may operate on a resource (e.g., a collection of information).

(182) The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented modules that operate to perform one or more operations or functions. The modules referred to herein may, in some example embodiments, comprise processor-implemented modules.

(183) Similarly, the methods or routines described herein may be at least partially processor-implemented. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented hardware modules. The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processor or processors may be located in a single location (e.g., within a home environment, an office environment or as a server farm), while in other embodiments the processors may be distributed across a number of locations.

(184) The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but deployed across a number of machines. In some example

embodiments, the one or more processors or processor-implemented modules may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example embodiments, the one or more processors or processor-implemented modules may be distributed across a number of geographic locations.

(185) Unless specifically stated otherwise, discussions herein using words such as “processing,” “computing,” “calculating,” “determining,” “presenting,” “displaying,” or the like may refer to actions or processes of a machine (e.g., a computer) that manipulates or transforms data represented as physical (e.g., electronic, magnetic, or optical) quantities within one or more memories (e.g., volatile memory, non-volatile memory, or a combination thereof), registers, or other machine components that receive, store, transmit, or display information.

(186) As used herein any reference to “one embodiment” or “an embodiment” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

(187) Some embodiments may be described using the expression “coupled” and “connected” along with their derivatives. For example, some embodiments may be described using the term “coupled” to indicate that two or more elements are in direct physical or electrical contact. The term “coupled,” however, may also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other. The embodiments are not limited in this context.

(188) As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

(189) In addition, use of the “a” or “an” are employed to describe elements and components of the embodiments herein. This is done merely for convenience and to give a general sense of the description. This description, and the claims that follow, should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise.

(190) This detailed description is to be construed as exemplary only and does not describe every possible embodiment, as describing every possible embodiment would be impractical, if not impossible. One may be implement numerous alternate embodiments, using either current technology or technology developed after the filing date of this application.

Claims

1. A computer-implemented method for facilitating using an autonomous vehicle as a payment device comprising: obtaining, by one or more processors, identification information for an autonomous vehicle, wherein the identification information for the autonomous vehicle includes a license plate number for the autonomous vehicle; detecting, by the one or more processors, a triggering event indicating that the autonomous vehicle is within a predetermined range for communicating with a merchant communication terminal; establishing, by the one or more processors, a short-range communication link between the merchant communication terminal and an electronic device proximately attached to the autonomous vehicle; receiving, by the one or more processors from the electronic device, a virtual order for goods or services provided by the merchant; transmitting, by the one or more processors to the electronic device, an electronic message requesting a user to authorize payment for the goods or services from the virtual account associated with the autonomous vehicle or the user; receiving, via the short-range communication link from the electronic device, input from the user authorizing payment from the goods or services from the virtual account; accepting, via the short-range communication link, the virtual order for goods or services; upon accepting the virtual order, automatically causing, by the one or more processors, the goods or services to be provided to the autonomous vehicle by controlling a device for providing the goods or services, wherein the goods or services are provided autonomously to the autonomous vehicle by directing the autonomous vehicle to a particular location for receiving the goods or services to control operation of the autonomous vehicle to

cause the autonomous vehicle to travel to the particular location; and transmitting, via the short-range communication link, a notification that electronic payment for the goods or service has been accepted.

2. The computer-implemented method of claim 1, wherein the goods or services provided autonomously by the merchant is gasoline and the merchant communication terminal is communicatively coupled to a smart gas pump having a robotic arm for maneuvering a nozzle to automatically pump gas into a gas tank opening of the autonomous vehicle.

3. The computer-implemented method of claim 2, further comprising: detecting, by the one or more processors, that a gas tank of the autonomous vehicle is open and a location of the gas tank opening for the autonomous vehicle with respect to the merchant communication terminal; directing, by the one or more processors, the robotic arm included in the smart gas pump to insert the nozzle into the gas tank opening according to the position of the gas tank opening; and automatically providing, by the one or more processors, gasoline to the autonomous vehicle.

4. The computer-implemented method of claim 1, further comprising: accepting, by the one or more processors, payment from the virtual account for the goods or services; and transmitting, via the short-range communication link, an electronic receipt for the payment of the goods or services.

5. The computer-implemented method of claim 1, wherein detecting the triggering event includes at least one of: detecting, by the one or more processors, that the autonomous vehicle is within Wi-Fi, Radio Frequency Identification (RFID), or Bluetooth communication range of the merchant communication terminal by receiving a Wi-Fi, RFID, or Bluetooth signal from the autonomous vehicle; or detecting, by the one or more processors, that the autonomous vehicle is within a predetermined distance of the merchant communication terminal by capturing images, via a camera communicatively coupled to the merchant communication terminal, of an area surrounding the merchant communication terminal, and identifying, using digital image analysis techniques, the autonomous vehicle within at least one of the images and a position of the autonomous vehicle relative to the merchant communication terminal.

6. A merchant communication terminal for facilitating using an autonomous vehicle as a payment device, the merchant communication terminal being associated with a merchant that provides goods or services, the merchant communication terminal comprising: one or more processors in a merchant communication terminal associated with a merchant; a non-transitory computer-readable memory coupled to the one or more processors and storing thereon instructions that, when executed by the one or more processors, cause the merchant communication terminal to: obtain identification information for an autonomous vehicle, wherein the identification information for the autonomous vehicle includes a license plate number for the autonomous vehicle; detect a triggering event indicating that the autonomous vehicle is within a predetermined range for communicating with the merchant communication terminal; establish a short-range communication link between the merchant communication terminal and an electronic device proximately attached to the autonomous vehicle; receive, from the electronic device, a virtual order for goods or services provided by the merchant; transmit, to the electronic device, an electronic message requesting a user to authorize payment for the goods or services from the virtual account associated with the autonomous vehicle or the user; receive, via the short-range communication link from the electronic device, input from the user authorizing payment from the goods or services from the virtual account; accept, via the short-range communication link, the virtual order for goods or services; upon accepting the virtual order, automatically cause the goods or services to be provided to the autonomous vehicle by controlling a device for providing the goods or services, wherein the goods or services are provided autonomously to the autonomous vehicle by directing the autonomous vehicle to a particular location for receiving the goods or services to control operation of the autonomous vehicle to cause the autonomous vehicle to travel to the particular location; and transmit, via the short-range communication link, a notification that electronic payment for the goods or service has been accepted.

7. The merchant communication terminal of claim 6, wherein the goods or services provided autonomously by the merchant is gasoline and the merchant communication terminal is communicatively coupled to a smart gas pump having a robotic arm for maneuvering a nozzle to automatically pump gas into a gas tank opening of the autonomous vehicle.

8. The merchant communication terminal of claim 7, wherein the instructions further cause the merchant communication terminal to: detect that a gas tank of the autonomous vehicle is open and a location of the gas tank opening for the autonomous vehicle with respect to the merchant communication terminal; direct the robotic arm included in the smart gas pump to insert the nozzle into the gas tank opening according to the position of the gas tank opening; and automatically provide gasoline to the autonomous vehicle.

9. The merchant communication terminal of claim 6, wherein the instructions further cause the merchant communication terminal to: accept payment from the virtual account for the goods or services; and transmit, via the short-range communication link, an electronic receipt for the payment of the goods or services.

10. The merchant communication terminal of claim 6, wherein to detect a triggering event indicating that an autonomous vehicle is within a predetermined range for communicating with the merchant communication terminal, the instructions further the merchant communication terminal to at least one of: detect that the autonomous vehicle is within Wi-Fi, Radio Frequency Identification (RFID), or Bluetooth communication range of the merchant communication terminal by receiving a Wi-Fi, RFID, or Bluetooth signal from the autonomous vehicle; or detect that the autonomous vehicle is within a predetermined distance of the merchant communication terminal by capturing images, via a camera communicatively coupled to the merchant communication terminal, of an area surrounding the merchant communication terminal, and identifying, using digital image analysis techniques, the autonomous vehicle within at least one of the images and a position of the autonomous vehicle relative to the merchant communication terminal.

11. A tangible, non-transitory computer-readable medium storing executable instructions for facilitating using an autonomous vehicle as a payment device, the instructions, when executed by one or more processors of a merchant communication terminal, cause the merchant communication terminal to: obtain identification information for an autonomous vehicle, wherein the identification information for the autonomous vehicle includes a license plate number for the autonomous vehicle; detect a triggering event indicating that the autonomous vehicle is within a predetermined range for communicating with the merchant communication terminal; establish a short-range communication link between the merchant communication terminal and an electronic device proximately attached to the autonomous vehicle; receive, from the electronic device, a virtual order for goods or services provided by the merchant; transmit, to the electronic device, an electronic message requesting a user to authorize payment for the goods or services from the virtual account associated with the autonomous vehicle or the user; receive, via the short-range communication link from the electronic device, input from the user authorizing payment from the goods or services from the virtual account; accept, via the short-range communication link, the virtual order for goods or services; upon accepting the virtual order, automatically cause the goods or services to be provided to the autonomous vehicle by controlling a device for providing the goods or services, wherein the goods or services are provided autonomously to the autonomous vehicle by directing the autonomous vehicle to a particular location for receiving the goods or services to control operation of the autonomous vehicle to cause the autonomous vehicle to travel to the particular location; and transmit, via the short-range communication link, a notification that electronic payment for the goods or service has been accepted.

12. The tangible, non-transitory computer-readable medium of claim 11, wherein the goods or services provided autonomously by the merchant is gasoline and the merchant communication terminal is communicatively coupled to a smart gas pump having a robotic arm for maneuvering a nozzle to automatically pump gas into a gas tank opening of the autonomous vehicle, the instructions further the merchant communication terminal to: detect that a gas tank of the autonomous vehicle is open and a location of the gas tank opening for the autonomous vehicle with respect to the merchant communication terminal; direct the robotic arm included in the smart gas pump to insert the nozzle into the gas tank opening according to the position of the gas tank opening; and automatically provide gasoline to the autonomous vehicle.

13. The tangible, non-transitory computer-readable medium of claim 11, wherein the instructions further cause the merchant communication terminal to: accept payment from the virtual account for the goods or services; and transmit, via the short-range communication link, an electronic receipt for the payment of the goods or services.

14. The tangible, non-transitory computer-readable medium of claim 11, wherein to detect a triggering event indicating that an autonomous vehicle is within a predetermined range for communicating with the merchant communication terminal, the instructions further the merchant communication terminal to at least one of: detect that the autonomous vehicle is within Wi-Fi, Radio Frequency Identification (RFID), or Bluetooth communication range of the merchant communication terminal by receiving a Wi-Fi, RFID, or Bluetooth signal from the autonomous vehicle; or detect that the autonomous vehicle is within a predetermined distance of the merchant communication terminal by capturing images, via a camera communicatively coupled to the merchant communication terminal, of an area surrounding the merchant communication terminal, and identifying, using digital image analysis techniques, the autonomous vehicle within at least one of the images and a position of the autonomous vehicle relative to the merchant communication terminal.
